RoSENet: Rotation and Similarity Enhancement Network for Multimodal Remote Sensing Image Land Cover Classification

Bokun Ma, Caihong Mu, *Senior Member, IEEE*, Yi Liu, Xinyu He, and Mosa Haidarh

***Abstract*— Multimodal classification methods have been widely applied in remote sensing (RS) land cover (LC) classification tasks. However, the existing multimodal classification methods often face challenges such as insufficient information extraction, sensitivity to noise, and inadequate utilization of complex spectral–spatial features during data fusion. To address these issues, this article proposes a rotation and similarity enhancement network (RoSENet) for multimodal RS LC classification by using hyperspectral image (HSI) data together with light detection and ranging (LiDAR) data. RoSENet consists of three key modules. First, the rotation fusion enhancement (RFE) module significantly increases the diversity of input HSI and LiDAR data and improves the model's generalization ability through multiangle rotation and spectral direction concatenation. Second, the spectral adaptive self-similarity convolution (SASSC) module ensures comprehensive preservation and fusion of spectral and spatial features through adaptive similarity measurement and convolution operations, enhancing the recognition of different classes in complex scenes. Third, the Euclidean similarity spatial-channel attention (ESSCA) module effectively strengthens global feature representation by capturing the resemblance between the central spectral vector and those situated in the surrounding area, improving the model's robustness in noisy environments. Extensive experiments are carried out on three public datasets, and the experimental results reveal that RoSENet demonstrates significant advantages in terms of overall accuracy (OA), average accuracy (AA), and the Kappa coefficient. Compared to traditional single-modal models, including convolutional neural networks (CNNs), and state-of-the-art single-modal and multimodal transformer models, RoSENet better captures detailed features in classification tasks and effectively reduces the impact of noise on classification accuracy.**

***Index Terms*— Convolutional neural networks (CNNs), hyperspectral image (HSI) classification, multimodal fusion, self-similarity.**

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Bokun Ma is with Guangzhou Institute of Technology, Xidian University, Guangzhou 510555, China (e-mail: bkma@stu.xidian.edu.cn).

Caihong Mu, Xinyu He, and Mosa Haidarh are with the School of Artificial Intelligence, Xidian University, Xi'an 710071, China (e-mail: mucaihongxd@foxmail.com; hexinyu@stu.xidian.edu.cn; mosahaidarh@stu.xidian.edu.cn).

Yi Liu is with the School of Electronic Engineering, Xidian University, Xi'an 710071, China (e-mail: yiliuxd@foxmail.com).

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I. Introduction

CLIMATE change and land cover (LC) change affect the ability of land to support human life, including biodiversity loss, ecosystem degradation, and climate change [1]. Therefore, in recent years, technologies for Earth observation (EO) through remote sensing (RS) have rapidly developed and been widely applied across various fields. These applications include, though not limited to, LC categorization [1], [2], [3], forestry management [4], mineral exploration and geographic mapping, object and target detection [5], [6], ecological surveillance [7], strategic urban development and planning [8], initiatives for preserving biodiversity, as well as crisis management and emergency response administration. Over the past few decades, these tasks have primarily relied on data from a single EO sensor. Hyperspectral imaging devices possess the capability to furnish extensive spectral and spatial data concurrently [9] and have thus been widely used.

Data from a single sensor often has limitations and may struggle to meet the growing accuracy requirements of real-world applications. Specifically, hyperspectral images (HSIs), which are widely used for LC classification, face numerous challenges. For example, the phenomena of spectral variability for the same object and spectral similarity for different objects commonly exist in HSIs. This leads to high intraclass variability and low interclass variability, making classification more difficult. Currently, classification is primarily carried out using convolutional neural network (CNN)-based deep learning methods, which typically rely on large amounts of training data. Nevertheless, in the context of practical applications of HSI classification, it is frequently challenging to meet the requirement of large training sets.

At the same time, LC classification using HSI data also faces challenges, such as data redundancy (i.e., similar spectral reflectance) and high noise levels in the collected spectral reflectance data. Due to the exceptionally high spectral resolution of HSI data, the correlation between adjacent bands is also high, resulting in significant data redundancy in the observed HSI data. This data redundancy makes it more complex to distinguish between different LC classes. Additionally, the high spectral resolution of hyperspectral sensors leads to lower energy in the light signals for each band, making the spectral reflectance signals more susceptible to various minor disturbances, resulting in a lower signal-to-noise ratio and the generation of more noise.

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To overcome the limitations of single-modal classification, multimodal LC classification (MLCC) has emerged as an important direction. The collection of different types of RS data enables the joint use of multimodal datasets to improve LC classification performance. The integration of data from disparate modalities can provide distinctive supplementary information, thereby facilitating the acquisition of a more comprehensive feature representation. As an illustration, light detection and ranging (LiDAR) technology is capable of capturing depth and elevation data, which facilitates precise elevation measurements. Such information is highly valuable and often absent in other data types, making LiDAR uniquely advantageous. From the perspective of RS imaging systems, spatial resolution and spectral resolution are often difficult to balance [10]. The high spectral resolution of HSI data typically restricts its spatial resolution, resulting in spectral mixing, which hinders LC classification in complex scenes [11]. HSI data, for instance, can effectively differentiate between water and grass, but it often struggles to distinguish roads from roofs when they are made of similar materials. In contrast, LiDAR provides elevation information, which can distinguish between objects with similar spectral characteristics but different elevations. Therefore, elevation information from LiDAR is highly beneficial for LC classification [12].

To address the insufficient utilization of complex relationships between spectral and spatial features and the sensitivity to noise in existing methods, thereby improving multimodal RS LC classification performance, we propose a novel network: rotation and similarity enhancement network (RoSENet). This network integrates HSI and LiDAR data by evaluating the similarity between spectral and spatial features and taking advantage of the elevation details obtained from LiDAR. Simultaneously, adaptive similarity metrics along with attention mechanisms are employed to accurately capture the intricate relationships between HSI and LiDAR data, thus boosting the classification performance through spatial feature extraction from both data types. The primary contributions presented in this article can be summarized as follows.

1) We propose a novel network named RoSENet for multimodal RS LC classification. By introducing the elevation information from LiDAR data, RoSENet employs adaptive similarity measurement and attention mechanisms to extract the complex relationships between HSI and LiDAR data, thereby improving classification performance.
2) To maximize the use of relational information between the central spectral vector and the spatially adjacent vectors within a patch, we propose a spectral adaptive self-similarity convolution (SASSC) module. By capturing self-similarity, this module better identifies detailed features of different classes, allowing the model to maintain high classification accuracy even in complex scenes. While extracting spatial features, it preserves the spectral characteristics of hyperspectral data through adaptive mechanisms, addressing the issue that traditional convolution operations potentially lose some spectral information when extracting spatial features.
3) We propose a Euclidean similarity spatial-channel attention (ESSCA) module, which enhances the global representation of spectral–spatial features by utilizing the relational information between the central feature vector and its spatial neighboring spectral feature vectors. This avoids relying solely on local information, thereby improving the ability of the model to understand complex scenes. The module dynamically adjusts the importance of each spectral channel and spatial location based on similarity, allowing the model to automatically focus on key regions for classification and reduce the impact of noise.
4) We conduct comprehensive experiments on three multimodal datasets, containing both HSI and LiDAR data. In these experiments, our method outperforms traditional machine learning methods, classic single-modal CNNs, and current state-of-the-art single-modal and multimodal transformer models in capturing detailed features in classification tasks while effectively reducing the impact of noise on classification accuracy. Additionally, we perform ablation studies using multimodal HSI and LiDAR data in order to illustrate the discrete contributions of the three modules and conduct parameter experiments to verify the stability of RoSENet under different settings.

We organize the remainder of this article as follows. Section II discusses traditional machine learning and deep learning methods for HSI classification, transformer-based methods, and self-attention methods. In Section III, we provide a detailed introduction of each component of the proposed RoSENet model. Then, in Section IV, we conduct extensive experiments to demonstrate and analyze the effectiveness of the proposed model. Finally, Section V concludes this article and proposes potential future research directions.

II. Related Work

A. Traditional Methods Based on Machine Learning and Deep Learning

Traditional machine learning methods have been widely applied in HSI classification, demonstrating effective results even with limited training samples [13], [14], [15]. Generally, these techniques involve two key stages. The first step is to represent HSI data within a feature space, aiming to reduce dimensionality and derive the most relevant features. Next, the derived features are used by a spectral classifier to perform the classification task [2], [16], [17], [18], [19]. Support vector machines (SVMs) and extreme learning machines (ELMs) are particularly noted for their superior performance in scenarios involving small sample sizes [20], [21]. However, the performance of traditional classifiers often relies on feature engineering, which requires manually designing appropriate feature extraction methods. As data complexity increases, traditional methods exhibit limitations, especially in capturing spatial information.

Later, deep learning technologies emerged, and researchers gradually introduced them into HSI classification tasks. Through multilayer neural network structures, deep learning methods could automatically and effectively extract spectral

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and spatial features, particularly excelling in multimodal data fusion [22]. For example, Hong et al. [23] proposed a novel technique for integrating HSI and LiDAR data, significantly improving classification accuracy. Additionally, CNN structures have been widely applied in HSI classification, including 1-D convolution (CNN1D), 2-D convolution (CNN2D), and 3-D convolution (CNN3D), which effectively capture spatial context information in HSI classification [24], [25], [26]. However, CNNs face limitations in capturing sequential features, particularly long-term dependencies, which results in diminished operational efficacy when confronted with classes exhibiting analogous spectral features.

B. Data Augmentation Methods

In HSI classification, the scarcity of labeled samples often leads to overfitting, and the data augmentation strategies have been widely used to increase the diversity of training samples without requiring additional labeled data [27]. These methods improve model robustness and classification performance, especially when labeled data is limited.

Li et al. [28] proposed the pixel block pair (PBP) method, which generated new training samples by pairing pixel blocks, significantly increasing the diversity and quantity of samples. Hao et al. [29] proposed a multilabel sample augmentation method, which assigned a single label to only a few pixels, while applying multilabel annotation to larger pixel regions, improving the classifier's accuracy. Wang et al. [30] introduced two paired data augmentation (PDA) methods and designed a soft loss-based Siamese CNN, which not only increased the diversity of training samples but also improved HSI classification accuracy by adaptively adjusting the loss function. Several studies tried to combine image rotation with HSI classification. Zheng et al. [31] proposed a rotation-invariant attention network (RIAN), which addressed performance degradation caused by image rotation in HSI classification by introducing a central spectral attention and corrected spatial attention module. Chen et al. [32] introduced a new Proto-MaxUp (PM) framework, which applied optimized data augmentation strategies to different data input parts (support set, query set, and new tasks) of the prototypical network (P-Net), reducing overfitting in the P-Net classifier. However, these methods also lead to models relying on specific augmented patterns and failing to fully learn the essential features of the original data, thereby reducing their fitting ability.

C. Multimodal LC Classification

In the study on LC classification, Khodadadzadeh et al. [33] utilized LiDAR data to derive elevation features. Many studies focus on utilizing the complementary information from HSIs and other multimodal data sources, such as using extended morphological profiles (EMPs) and attribute profiles (APs) containing elevation information, to jointly extract and classify features from HSI and LiDAR data through methods based on SVM or random forest (RF) [11], [34], [35].

Rasti et al. [36] enhanced the process of joint extraction of EMPs through the utilization of total variation component analysis and employed SVM and RF as classifiers for the fusion of features. Merentitis et al. [37] proposed an automated fusion technique utilizing an RF classifier for joint classification of HSI and LiDAR data. However, these traditional methods struggle to fully exploit spectral and spatial information and are poorly adaptive to data redundancy and noise, limiting the model's performance.

In response, Li et al. [38] proposed a fusion model based on 1-D and 2-D CNNs for spectral and spatial feature extraction, and designed a cascade net for LiDAR feature extraction. However, CNNs are not effective at capturing long-range dependencies, whereas vision transformer (ViT) can better address this issue by using a self-attention mechanism. ViT not only captures long-range structures and patterns but also integrates spatial and positional information, improving the accuracy of LC classification.

Consequently, Roy et al. [39] proposed a multimodal fusion transformer (MFT), which combined HSI and LiDAR data and used class labels obtained from LiDAR data to interact with HSI sub-patch labels. This interaction delivers spectral–spatial information through attention mechanisms, which in turn improves classification performance. Although MFT represents a significant breakthrough, the method still faces challenges with insufficient information fusion due to the differences between LiDAR and HSI modalities and the limitations of the attention mechanism in cross-modal interaction.

D. Methods Based on Self-Attention Mechanism

To overcome the limitations of traditional deep learning methods, which often neglect the global relationships between spectral and spatial dimensions when capturing complex spectral–spatial features, the significance of transformer models in HSI classification has been progressively recognized. Unlike CNNs, transformer models, due to their self-attention mechanism, excel at processing and analyzing sequential data, which has been causing great concern [40]. This mechanism not only effectively captures the global associations of features but also handles complex information better. Inspired by the success of transformers in natural language processing and computer vision, researchers have begun developing transformer models suitable for HSI classification [41], [42]. For instance, Hong et al. [15] proposed SpectralFormer, a transformer model that leveraged information from adjacent spectral bands to learn spectral representations, significantly improving the accuracy of HSI classification. Additionally, Sun et al. [43] introduced the spectral–spatial feature tokenization (SSFT) Transformer model, which further improved classification performance through the combination of local spatial semantic information with relationships between sequences. However, the main challenge of SSFT lay in its large number of parameters, limiting its application in multimodal data fusion. Zhou et al. [44] proposed a dual-branch convolutional transformer network (DCTN) with efficient interactive self-attention (EISA) for HSI

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classification, effectively combining the strengths of CNNs and Transformers to capture both local and global features, and outperforming existing methods across multiple datasets. Li et al. [45] proposed an SNN framework for hyperspectral classification, integrating an adaptive refocusing convolutional layer that dynamically adjusts kernel responses and a spike self-attention mechanism that models long-range dependencies through membrane potential accumulation, thereby enhancing feature extraction and classification accuracy while maintaining low computational energy consumption. These methods combining CNNs and Transformers have achieved good results on unimodal datasets, which also provides new ideas for model construction on multimodal datasets.

Subsequently, to further enhance the capabilities of transformer models, researchers began to combine transformers with multimodal data. Ding et al. [46] developed a global-local transformer network (GLT-Net), which integrated the long-range dependency modeling capabilities of transformers with the local feature representation abilities of convolutional operators, effectively improving the joint classification performance of HSI and LiDAR data. Zhao et al. [47] proposed a dual-branch network composed of a transformer and hierarchical CNN, which fused multisource heterogeneous information for classification. This network significantly improved classification performance by combining self-attention mechanisms with feature-level fusion of multimodal data. Yu et al. [48] introduced the cross-context capsule ViT (CapViT), which effectively enhanced LC classification accuracy on MS-LiDAR data by capturing interactions between features at multiple scales and incorporating long-range dependencies.

To further improve the performance of transformer models in multimodal data fusion, researchers also developed models such as the local information interaction transformer (LIIT) [49] and the multiattention hierarchical dense fusion network (MAHiDFNet) [22], which enhanced cross-modal information interaction through attention mechanisms, significantly improving model classification performance. Roy et al. [39] proposed an MFT network based on a multihead cross-subpatch attention mechanism, resulting in effective fusion of multisource data and improvement of RS image classification performance. Li et al. [50] significantly enhanced the accuracy and efficiency of HSI classification by introducing multiple self-similarity and spectral–spatial feature fusion modules. However, for self-attention-based methods, the feature differences between LiDAR and HSI modalities and the limitations of attention mechanisms in cross-modal interaction result in insufficient information fusion.

In summary, traditional machine learning methods perform well on simple data, achieving efficient classification through manually designed feature extraction methods together with a classifier. However, they exhibit limitations when dealing with complex data. By developing technologies for deep learning, deep neural networks, especially CNNs, have made significant progress in HSI classification, although CNNs have certain limitations in capturing sequential features. To address the issue of insufficient samples in HSI classification, researchers have proposed various data augmentation methods that increase sample diversity to improve model classification performance. Additionally, transformer models based on self-attention mechanisms have gradually become a research hotspot in HSI classification due to their advantages in handling sequential data. By integrating global and local information, transformer models effectively improve classification accuracy and are increasingly applied in multimodal data fusion. However, these models still face challenges such as a large number of parameters and insufficient information fusion between multimodal data. Researchers have further optimized HSI and LiDAR joint classification performance by developing new transformer network structures.

Based on the analysis of the strengths and weaknesses of the aforementioned studies, this article proposes RoSENet, a novel network that integrates LiDAR and HSI data to improve LC classification performance.

III. Methodology

As Fig. 1 shows, the proposed RoSENet is a multimodal classification model based on rotation and similarity enhancement. This network utilizes HSI and LiDAR data, which passes through three different modules, including rotation fusion enhancement (RFE), SASSC, and ESSCA, achieving high classification accuracy. Sections III-A–III-C describe the proposed multimodal deep learning framework step by step.

A. Rotation Fusion Enhancement

This article proposes an HSI and LiDAR data enhancement module based on multiangle rotation and multimodal fusion to improve the model's generalization capacity and effectiveness in classification tasks. This module increases the diversity of training samples by performing rotation transformations on image patches from HSI and LiDAR images and concatenating them along the spectral dimension. RFE performs small-angle rotations, which enables the model to extract more detailed directional semantic information for the subsequent SASSC and ESSCA modules. A detailed description is provided below.

In situations where training samples are limited, HSI and LiDAR data may exhibit significant directional differences under different viewing angles. To address this issue, we propose a rotation and concatenation method based on image patches. Specifically, for each input HSI and LiDAR data sample, we first extract image patches of size $h \times w$, and rotate these patches around their center by $\mathcal{K}$ predefined angles denoted by θ_k, which belongs to $(0°, 45°, 90°, 135°)$. We compute the rotated image patches using the following rotation matrix:

$$\begin{aligned}\begin{pmatrix} i' \\ j' \end{pmatrix} &= \mathcal{R}\left(\begin{pmatrix} i \\ j \end{pmatrix}, \theta_k\right) \\ &= \begin{pmatrix} \cos\theta_k & -\sin\theta_k \\ \sin\theta_k & \cos\theta_k \end{pmatrix}\begin{pmatrix} i - i_c \\ j - j_c \end{pmatrix} + \begin{pmatrix} i_c \\ j_c \end{pmatrix} \\ i_c &= \left\lfloor \frac{h-1}{2} \right\rfloor, \quad j_c = \left\lfloor \frac{w-1}{2} \right\rfloor \end{aligned} \tag{1}$$

where $\mathcal{R}(\cdot)$ is a rotation operation that takes the original coordinates (i, j) and the rotation angle θ_k as input and outputs the rotated pixel coordinates (i', j'). The values (i_c, j_c)

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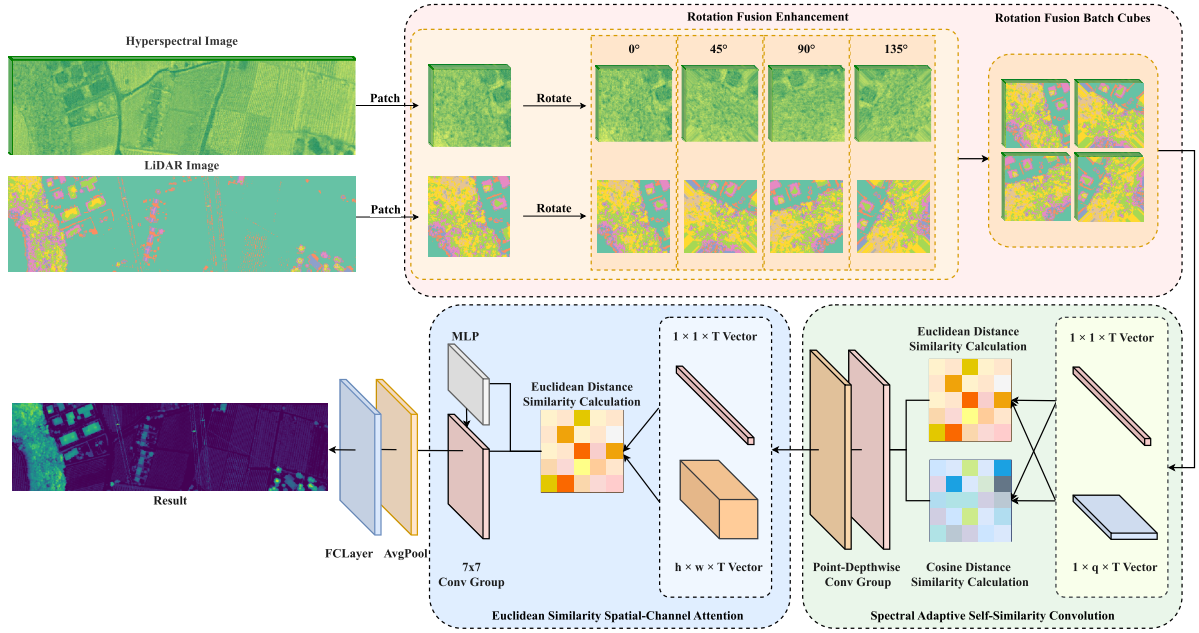


Fig. 1. Flowchart of the proposed RoSENet for HSI and LiDAR data fusion. First, the RFE module increases data diversity with multiangle rotations; next, the SASSC module integrates spectral and spatial features; finally, the ESSCA module captures local and global dependencies, all working together to enhance classification accuracy in multimodal fusion tasks.

represent the center coordinates of the image patch. The rotated pixel coordinates (i', j') are then mapped back to the original grid, generating the rotated image patches $\mathbf{Z}^{\mathrm{H}}_{n,\theta_k}$ (where H represents HSI) and $\mathbf{Z}^{\mathrm{L}}_{n,\theta_k}$ (where L represents LiDAR) based on the original image patches $\mathbf{Z}^{\mathrm{H}}_{n}$ and $\mathbf{Z}^{\mathrm{L}}_{n}$. The formulas are as follows:

$$\begin{aligned}\mathbf{Z}^{\mathrm{H}}_{n,\theta_k}(i', j', :) &= \mathcal{R}\left(\mathbf{Z}^{\mathrm{H}}_{n}(i, j, :), \theta_k\right)\\ \mathbf{Z}^{\mathrm{L}}_{n,\theta_k}(i', j', :) &= \mathcal{R}\left(\mathbf{Z}^{\mathrm{L}}_{n}(i, j, :), \theta_k\right)\end{aligned} \tag{2}$$

where $\mathbf{Z}^{\mathrm{H}}_{n}(i, j, :)$ and $\mathbf{Z}^{\mathrm{L}}_{n}(i, j, :)$ represent the vector at the coordinates (i, j) in the nth HSI and LiDAR patch. Here, $n \in N$ is used to identify the index of each input sample, with N representing the total number of samples.

Through this rotation operation, the image patches can capture spatial information from different directions, thereby reducing the impact of absolute directionality on model training. Subsequently, the rotated HSI and LiDAR image patches are concatenated along the spectral dimension, generating the concatenated image patches $\mathbf{Z}^{\mathrm{C}}_{n,\theta_k}$, and constructing the final enhanced dataset. The mathematical expression of this process is as follows:

$$\begin{aligned}\mathbf{Z}^{\mathrm{C}}_{n,\theta_k} &= \mathcal{C}\left(\mathbf{Z}^{\mathrm{H}}_{n,\theta_k}, \mathbf{Z}^{\mathrm{L}}_{n,\theta_k}\right)\\ \mathcal{X}^{\mathrm{aug}} &= \left\{\mathbf{Z}^{\mathrm{C}}_{n,\theta_k}\right\}_{n=1}^{N}, \quad \theta_k \in \{\theta_1, \theta_2, \ldots, \theta_K\}\end{aligned} \tag{3}$$

where $\mathcal{C}(\cdot)$ represents the concatenation of HSI and LiDAR image patches along the spectral dimension, resulting in the concatenated image patches $\mathbf{Z}^{\mathrm{C}}_{n,\theta_k}$ with dimensions $h \times w \times (B_{\mathrm{H}} + B_{\mathrm{L}})$, where B_{H} and B_{L} represent the spectral dimensions (number of bands) of the HSI and LiDAR data. $\mathcal{X}^{\mathrm{aug}}$ is the enhanced dataset containing all the image patches after rotation and concatenation.

Through the enhancement and fusion by the RFE module, the model can more comprehensively learn information from different directions and different modalities, laying the foundation for the subsequent further fusion operation by SASSC, thereby effectively improving classification performance in HSI and LiDAR data fusion tasks. Experimental results show that this method has achieved significant performance improvements in multiple classification tasks involving the fusion of HSI and LiDAR data.

B. Spectral Adaptive Self-Similarity Convolution

The SASSC module is designed to further fuse HSI and LiDAR data and enhance the model's ability to capture spatial–spectral features. For high-dimensional data consisting of HSI and LiDAR, each band contains different semantic information. Cosine similarity is highly effective in capturing semantic similarity by measuring the directional consistency between these high-dimensional vectors [51]. In contrast, Euclidean distance directly calculates the geometric distance between samples and is suitable for distance measurement in feature space, making it effective in spatial feature extraction when using a Euclidean distance-based similarity attention mechanism [52]. However, Euclidean distance is scale-sensitive, as it directly reflects the differences between feature vectors [53]. Therefore, we adopt both similarity measurements with different properties when designing the SASSC, adaptively balancing the weights of them. SASSC fully extracts and utilizes the semantic information from HSIs and LiDAR data, achieving a balance between retaining the spectral characteristics of hyperspectral data and extracting spatial features. Fig. 2 illustrates the specific structure of this module. This module enhances feature learning through adaptive similarity measurement and convolution operations.

The SASSC module first receives the concatenated image patches generated by the RFE module as input. As mentioned earlier, the enhanced dataset of concatenated image patches is

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denoted as $\mathcal{X}^{\text{aug}} = \{\mathbf{Z}^{\text{C}}_{n,\theta_k}\}_{n=1}^{N}, \theta_k \in \{\theta_1, \theta_2, \ldots, \theta_K\}$, where $\mathbf{Z}^{\text{C}}_{n,\theta_k}$ represents the spectral–spatial data after multiangle rotation and concatenation. These concatenated image patches contain feature information from both HSI and LiDAR data at different angles, providing the model with rich training samples.

We first flatten the 3-D spectral image patch $\mathbf{Z}^{\text{C}}_{n,\theta_k}$ of size $h \times w \times T$ into a 2-D matrix of size $(h \times w) \times T$. Here, h and w represent the height and width of the patch, respectively, and $T = (B_{\text{H}} + B_{\text{L}})$ is the number of spectral bands. The flattening operation follows a row-major order, where the spatial dimensions h and w are combined into a single dimension, resulting in a matrix with $(h \times w)$ rows and T columns.

Specifically, the spectral vector $\hat{x}_{i,j}$ is the flattened version of the original spectral vector $x_{i,j}$, which is originally located at row i and column j of a concatenated image patch generated by RFE. This transformed vector becomes the qth element in the flattened 2-D matrix, where $q = i \times w + j$, with $i \in [0, h-1]$ and $j \in [0, w-1]$. Therefore, the index q represents the position of the vector within the flattened 2-D matrix, and $q \in \{0, 1, \ldots, h \times w - 1\}$.

In the original image patch $\mathbf{Z}^{\text{C}}_{n,\theta_k}$, the central spectral vector $c_{i,j} \in \mathbb{R}^T$ with the center coordinates (i_c, j_c) of the image patch is selected as the representative feature of the patch. To better understand and utilize this spectral information, the module calculates the Euclidean distance similarity matrix $S^E_{i,j}$ and the cosine similarity matrix $S^C_{i,j}$, which measure the similarity between c and its spatial neighboring spectral vectors, respectively.

Specifically, the Euclidean distance similarity matrix is calculated using the following formula:

$$S^E_{i,j} = \text{EDSim}\big(c, \mathbf{Z}^{\text{C}}_{n,\theta_k}\big) = \left[\frac{1}{1 + \|c - \hat{x}_q\|}\right]_q \tag{4}$$

where $\hat{x}_q$ represents any flattened spectral vector, and $q \in \{0, 1, \ldots, h \times w - 1\}$. The matrix $S^E_{i,j}$ captures the similarity according to the Euclidean distance between the central spectral vector c and each spectral vector, thereby reflecting the spatial–spectral feature differences. The notation $[\cdot]_q$ represents the computation of similarity for all query vectors.

Similarly, the cosine similarity matrix $S^C_{i,j}$ is calculated using the following formula:

$$S^C_{i,j} = \text{CosSim}\big(c, \mathbf{Z}^{\text{C}}_{n,\theta_k}\big) = \left[\frac{c \cdot \hat{x}_q}{\|c\|\|\hat{x}_q\|}\right]_q \tag{5}$$

where $S^C_{i,j}$ measures the similarity matrix between c and each spectral vector through the cosine angle, providing a direction-based similarity measurement method.

Next, to generate the self-similarity attention maps, the module normalizes the aforementioned similarity matrices $S^E_{i,j}$ and $S^C_{i,j}$ using the softmax function. The Euclidean self-similarity attention map $A^E_{i,j}$ and the cosine self-similarity attention map $A^C_{i,j}$ are calculated using the following formulas, respectively,

$$A^E_{i,j} = \text{Softmax}\big(S^E_{i,j}\big) = \left[\frac{\exp(e_q)}{\sum_{q=0}^{h\times w-1}\exp(e_q)}\right]_q \tag{6}$$

$$A^C_{i,j} = \text{Softmax}\big(S^C_{i,j}\big) = \left[\frac{\exp(c_q)}{\sum_{q=0}^{h\times w-1}\exp(c_q)}\right]_q \tag{7}$$

where e_q and c_q represent the corresponding qth element in $S^E_{i,j}$ and $S^C_{i,j}$.

Through these self-similarity attention maps, the module can more accurately capture the similarity patterns between the central spectral vector and those situated in the surrounding area. To further enhance the spatial representation of features, the module adopts an adaptive weighted fusion method. Initially, the trainable parameter λ is set to 0.5, which can be adjusted during training, combining the two self-similarity attention maps $A^E_{i,j}$ and $A^C_{i,j}$ to generate the weighted fused self-similarity attention map $A^\lambda_{i,j}$

$$A^\lambda_{i,j} = \lambda \times A^E_{i,j} + (1 - \lambda) \times A^C_{i,j}. \tag{8}$$

Finally, through element-wise multiplication and addition operations, the recalibrated spectral–spatial feature map $F'_{i,j}$ is generated

$$F'_{i,j} = A^\lambda_{i,j} \otimes \mathbf{Z}^{\text{C}}_{n,\theta_k} + \mathbf{Z}^{\text{C}}_{n,\theta_k} \tag{9}$$

where $\otimes$ represents the element-wise multiplication operation.

At this stage, $F'_{i,j}$ already contains adaptively weighted spectral–spatial information and serves as the basis for further processing. To extract more efficient features, the module subsequently processes $F'_{i,j}$ through 1×1 and 3×3 convolution layers. First, $F'_{i,j}$ is processed through batch normalization (BN) and a 1×1 convolution layer, combined with a LeakyReLU activation function, collectively referred to as the Conv 1×1 Group, to generate the intermediate feature $F''_{i,j}$

$$F''_{i,j} = \text{LeakyReLU}\big(W_{1\times1} * \text{BN}\big(F'_{i,j}\big)\big) \tag{10}$$

where $W_{1\times1}$ represents the weight of 1×1 convolution layer. Then, a 3×3 convolution layer with a ReLU activation function is applied to the intermediate feature $F''_{i,j}$, collectively referred to as the Conv 3×3 Group, resulting in the final output feature $F'''_{i,j} \in \mathbb{R}^{h\times w\times T}$

$$F'''_{i,j} = \text{ReLU}\big(W_{3\times3} * F''_{i,j} + \beta\big) \tag{11}$$

where $W_{3\times3}$ represents the weights of 3×3 convolution layer, and β represents the bias of 3×3 convolution layer.

The SASSC module effectively enhances the model's ability to further fuse HSI and LiDAR data, improving the capacity to capture spatial–spectral features. Experimental results show that this module exhibits significant performance improvements across various fusion tasks, validating its superiority in processing complex multimodal data.

C. Euclidean Similarity Spatial-Channel Attention

The ESSCA module is proposed to enhance the spectral–spatial feature representation of the fused HSI

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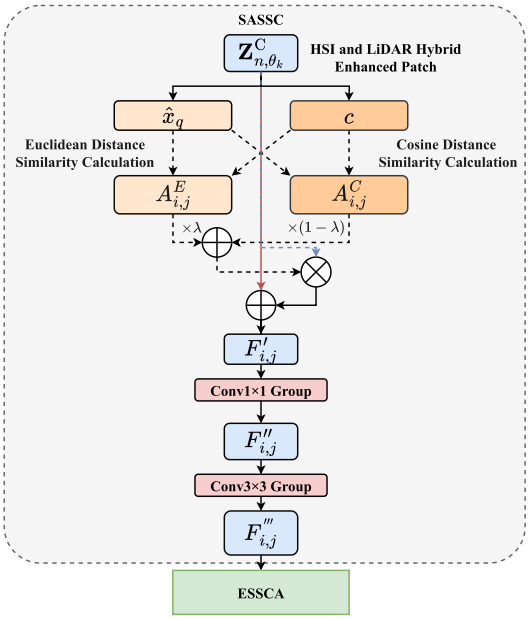


Fig. 2. Structural diagram of the proposed SASSC module.

and LiDAR data by precisely capturing both local and global dependencies in the feature space. After incorporating the LiDAR image, more spatial information is introduced compared to only using the HSI image. Once the fused high-level semantic information is extracted through the preprocessing steps, including RFE and SASSC, we only need to leverage the Euclidean distance in ESSCA as the similarity measurement, which is sensitive to the positional information between vectors, further enhancing the extraction of spatial information [54]. As Fig. 3 shows, the module is made up of three main components: Euclidean distance similarity computation, channel attention, and spatial attention.

The ESSCA module receives the feature map $\mathbf{X} = F'''_{i,j} \in \mathbb{R}^{h\times w\times T}$ from the preceding SASSC module as input. First, the feature map passes through a 1×1 convolution layer to generate the query matrix Q, key matrix K, and value matrix V, each with dimensions $h \times w \times T$.

To capture the relationships between local features, the central feature vector $\hat{x}_{\text{center}}$ is extracted from Q and expanded to match the dimension of K. Next, by calculating the Euclidean distance between the central feature vector $\hat{x}_{\text{center}}$ and the spatial neighboring spectral feature vectors (the vectors within K), the similarity matrix E_{dist} is obtained

$$E_{\text{dist}} = \frac{1}{1 + \|\hat{x}_{\text{center}} - K\|} \tag{12}$$

where $\|\cdot\|$ denotes the calculation of Euclidean distance. Subsequently, the similarity matrix E_{dist} is normalized using the softmax function to generate the self-similarity attention map α_{ED}

$$\alpha_{\text{ED}} = \text{Softmax}(E_{\text{dist}}). \tag{13}$$

Next, the value matrix V is weighted using the attention map α_{ED}, resulting in the calibrated feature map V'. Finally, the calibrated feature map V' is element-wise added to the input feature map $\mathbf{X}$, generating the updated feature map $\mathbf{X}'$

$$\begin{aligned} V' &= \alpha_{\text{ED}} \otimes V \\ \mathbf{X}' &= V' + \mathbf{X}. \end{aligned} \tag{14}$$

To further optimize the feature representation, the ESSCA module introduces adaptive channel and spatial attention mechanisms. First, the feature map $\mathbf{X}'$ undergoes channel attention processing, which is implemented using a multilayer perceptron (MLP) structure, generating the channel attention map M_c. Then, M_c is multiplied by the feature map $\mathbf{X}'$, resulting in the channel feature map $\mathbf{X}''$

$$\begin{aligned} M_c &= \sigma(W_2 \cdot \text{ReLU}(W_1 \cdot \mathbf{X}')) \\ \mathbf{X}'' &= M_c \otimes \mathbf{X}' \end{aligned} \tag{15}$$

where W_1 and W_2 are the weight matrices of the fully connected layers, and σ denotes the sigmoid activation function. The generated channel attention map M_c is used to perform element-wise weighting on the feature map, enhancing the dependency in the channel dimension.

Next, the feature map $\mathbf{X}''$ is processed through the spatial attention mechanism, generating the spatial attention map M_s. The map M_s is then multiplied by the feature map $\mathbf{X}''$, producing the final spectral-spatial feature map $\mathbf{X}'''$

$$\begin{aligned} \phi &= \text{BN}_1\left(\text{Conv2d}_1\left(\mathbf{X}''\right)\right) \\ \psi &= \text{Conv2d}_2(\text{ReLU}(\phi)) \\ M_s &= \sigma(\text{BN}_2(\psi)) \\ \mathbf{X}''' &= M_s \otimes \mathbf{X}'' \end{aligned} \tag{16}$$

where Conv2d_1 and Conv2d_2 are two 7×7 convolution layers (Conv 7×7 Group), and BN_1 and BN_2 are the corresponding batch normalization layers in 7×7 convolution group. The intermediate results are denoted by ϕ and ψ, representing the result after the first convolution and BN, and that after ReLU activation and the second convolution, respectively.

Through the above process, the ESSCA module effectively captures and utilizes the complex relationships within hyperspectral and LiDAR data. As a result, the module delivers exceptional performance in a variety of complex multimodal data fusion tasks, enhancing the model's ability to comprehensively represent spectral–spatial features.

IV. Experiments

A. Multimodal Datasets

This section considers three distinct HSI datasets and their associated multimodal data sources (LiDAR) to evaluate the performance of the proposed RoSENet network for multimodal RS LC classification. The experimental datasets include MUUFL Gulfport (MG), Trento, and the University of Houston (UH). For the MG dataset, we use a random sampling of 5% training samples, consistent with MFT [39], to ensure a fair comparison of our model with others, while for the Trento and UH datasets, we use standard training and test sample sets [55], [56].

1) *MG:* This scene was collected in November 2010 over the University of Southern Mississippi campus using the reflective optics system imaging spectrometer (ROSIS)

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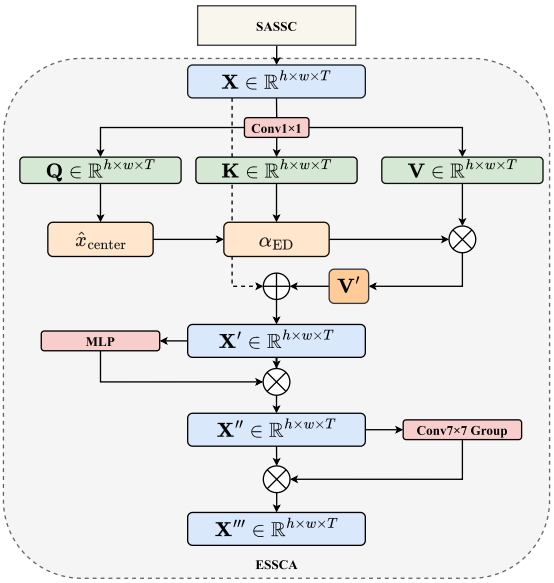


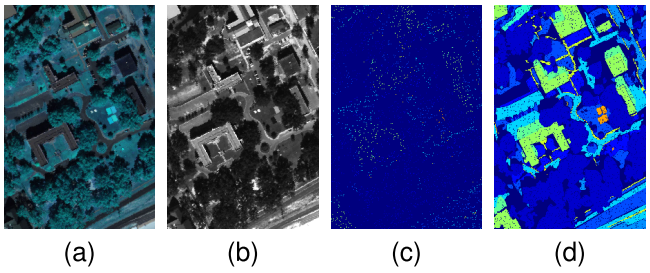
Fig. 3. Structural diagram of the proposed ESSCA module.

sensor [57], [58]. The HSI contains 325 × 220 pixels and 72 spectral bands. The LiDAR image contains two rasters of elevation data. Due to noise, the first eight and last eight bands were removed, leaving a total of 64 bands. The 64 retained bands still contain considerable noise, which increases the classification difficulty of the dataset. The data describe 11 urban LC categories, with a total of 53 687 ground-truth pixels. Fig. 4 shows the distribution of 5% randomly selected samples from each category.

2) *Trento:* Collected from the south rural area of Trento, Italy, the HSI data of it was captured by the AISA Eagle sensor, and the LiDAR data of it was collected by the Optech ALTM 3100EA sensor. Each HSI contains 63 bands, covering a wavelength range from 0.42 to 0.99 μm, and the LiDAR dataset consists of a single raster that provides elevation information. The spectral resolution achieved is 9.2 nm, and pixels of six non-overlapping vegetation LC types. Information on the training and test samples for each category is shown in Fig. 5.

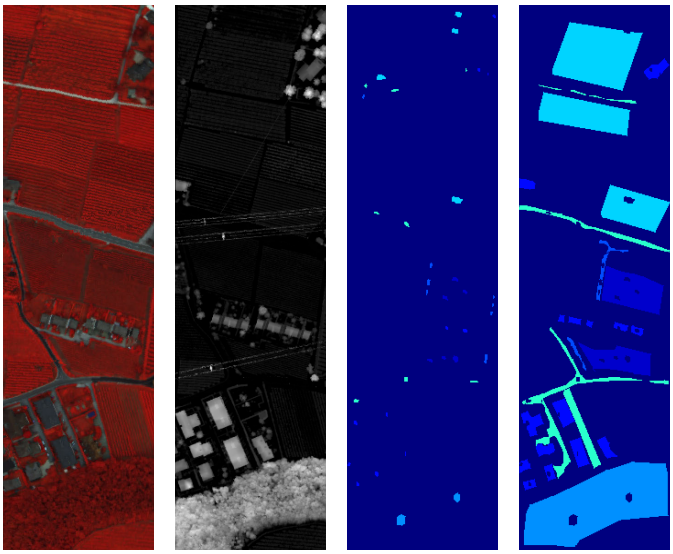
3) *UH:* The scene was acquired using a compact airborne spectrographic imager (CASI) and made available by the IEEE Geoscience and Remote Sensing Society as part of the 2013 data fusion contest. In our experiments, we utilized the HSI and LiDAR images. Each image has dimensions of 340 × 1905 pixels, with the HSI consisting of 144 spectral bands ranging from 0.38 to 1.05 μm. The dataset has a spatial resolution of 2.5 m/pixel. The ground truth consists of 15 different LC categories. The samples for the 15 LC categories are separated into training and test samples of fixed size. Fig. 6 shows the 15 distinct LC classes along with their corresponding training and test samples.

Figs. 4–6 provide the category-specific summaries for each scene, respectively. Each dataset contains its corresponding ground truth, the types related to LC categories, and the number of labeled samples available for each category. As shown in Figs. 4–6, the partition of training and test sets of MG is different from those of Trento and UH. The former is traditional random sampling, while the latter is non-overlapping sampling, where the training samples and test samples are spatially non-overlapping and the difficulty of classification is increased.



Color	No.	Land cover	Train	Test	Color	No.	Land cover	Train	Test
	0	Background	68817	20496		1	Trees	1162	22084
	2	Grass-Pure	214	4056		3	Grass-Groundsurface	344	6538
	4	Dirt-And-Sand	91	1735		5	Road-Materials	334	6353
	6	Water	23	443		7	Buildings'-Shadow	112	2121
	8	Buildings	312	5928		9	Sidewalk	69	1316
	10	Yellow-Curb	9	174		11	ClothPanels	13	256

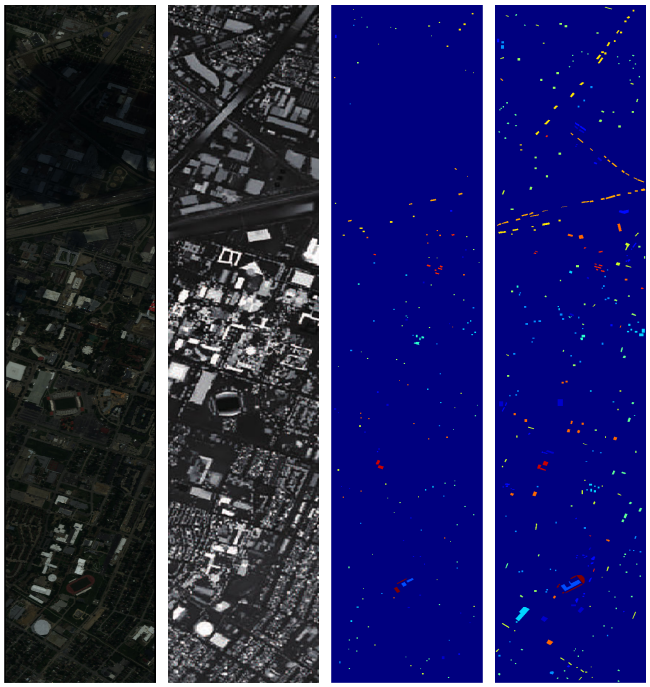
Fig. 4. MG scene. (a) Color composite illustration of the HSI, created using bands 40, 20, and 10 as R, G, and B. (b) Grayscale image generated from the LiDAR data. (c) Ground truth of the training samples for the MG scene. (d) Ground truth of the test samples for the MG scene. The table displays the category-specific LC types, along with the number of randomly selected 5% training samples and the remaining 95% test samples.



Color	No.	Land cover	Train	Test	Color	No.	Land cover	Train	Test
	0	Background	98781	70205		1	Apples	129	3905
	2	Buildings	125	2778		3	Ground	105	374
	4	Woods	154	8969		5	Vineyard	184	10317
	6	Roads	122	3052					

Fig. 5. Trento scene. (a) Color composite illustration of the HSI, created using bands 40, 20, and 10 as R, G, and B. (b) Grayscale image generated from the LiDAR data. (c) Ground truth of the non-overlapping training samples. (d) Ground truth of the non-overlapping test samples. The table displays the category-specific LC types, along with the number of non-overlapping training and test samples for each category.

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Color	No.	Land cover	Train	Test	Color	No.	Land cover	Train	Test
	0	Background	662013	652648		1	Grass-healthy	198	1053
	2	Grass-stressed	190	1064		3	Grass-synthetic	192	505
	4	Tree	188	1056		5	Soil	186	1056
	6	Water	182	143		7	Residential	196	1072
	8	Commercial	191	1053		9	Road	193	1059
	10	Highway	191	1036		11	Railway	181	1054
	12	Parking-lot1	192	1041		13	Parking-lot2	184	285
	14	Tennis-court	181	247		15	Running-track	187	473

Fig. 6. UH scene. (a) Color composite illustration of the HSI, created using bands 64, 43, and 22 as R, G, and B. (b) Grayscale image generated from the LiDAR data. (c) Ground truth of the non-overlapping training samples. (d) Ground truth of the non-overlapping test samples. The table displays the category-specific LC types, along with the number of non-overlapping training and test samples for each category.

B. Experimental Setup

To validate the effectiveness of the RoSENet model being proposed, we designed and implemented extensive experiments, comparing it with traditional methods and the latest techniques. The traditional methodologies compared comprise classic classifiers such as KNN [59], RF [2], and SVM [20], and conventional CNN models, including CNN1D [24], CNN2D [25], and CNN3D [26]. Additionally, we compared it with some cutting-edge Transformer models, such as ViT [41], SpectralFormer [15], and MFT [39]. Among these comparison methods, MFT is a multimodal method, while the other methods are all unimodal. All methods in the experiments were performed using both HSI and LiDAR data. Notably, in single-modal approaches, the HSI data and LiDAR data were concatenated by band.

All experiments were conducted on a Windows 11 system with high-end hardware. The setup included an **Intel Core i7-10700** processor (eight cores, 16 threads), **16 GB** RAM for efficient memory management, and an **NVIDIA GeForce GTX 1660 SUPER GPU (6 GB VRAM)** for graphics processing and deep learning acceleration. This configuration ensured smooth performance, efficient data processing, and reliable model validation.

During the experiments, we set the model's **batch sizes** to 64 (for training) and 500 (for testing) to ensure the stability and efficiency of the model at different stages. The patch size extracted from the HSI data was $11 \times 11 \times B_H$, a size determined through extensive testing to effectively capture spectral features without losing spatial information. Meanwhile, the patch size extracted from the LiDAR data was $11 \times 11 \times B_L$, which helped more accurately represent terrain features. The proposed RoSENet model was optimized using the **AdamW optimizer**, which has been validated for its excellent weight regularization in various tasks [60], [61]. Additionally, a cosine annealing learning rate scheduler with a maximum number of iterations set to 20 and a minimum learning rate of $1e^{-6}$ was employed to dynamically adjust the rate of learning as the training progresses, further enhancing the model's convergence. In comparison experiments, except for KNN, RF, SVM, and RoSENet, all other comparison models were trained using the **Adam optimizer**, with a precise **learning rate** of $5e^{-4}$ and the weight decay parameter of $5e^{-3}$ to balance the model's learning speed and prevent overfitting. To further enhance training efficiency, these models using the Adam optimizer were also equipped with a step scheduler, where the step size was set to 50 and the decay factor was 0.9. This setup allowed dynamic adjustment of the learning rate during training, facilitating better convergence within the 200 **training epochs**. To ensure the reliability of the experimental results, each experiment was repeated ten times, with the average and standard deviation of all experiments reported to provide a comprehensive evaluation of model performance. It is worth noting that PyTorch 2.1.1 and Python 3.10 were used to develop all RoSENet source code, ensuring stability and efficiency while establishing a reliable benchmark for future research.

Evaluation metrics and implementation details: To evaluate the performance of the proposed model, we adopted several widely used metrics: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient (κ). OA measures the proportion of correctly classified samples, while AA focuses on the classification accuracy of each class. The Kappa coefficient measures the consistency between the predicted classification and ground truth, and a value closer to 1 indicates higher consistency. The experiments were conducted under two configurations, as follows.

1) In Sections IV-C–IV-F, the MG dataset used 5% random sampling for training, and the UH and Trento datasets used independent training and test sets, ensuring no overlap. This setup tests the model's generalization ability and performance in a strictly separated data environment.
2) In Section IV-G, random sampling was applied to all three datasets, analyzing the model's performance in a conventional scenario.

C. Comparison Experiments

Quantitative metrics for the proposed model and other methods using HSI and LiDAR data of the MG dataset are presented in Table I. The metrics include OA, AA, Kappa

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TABLE I

COMPARISON OF OA, AA, AND KAPPA (κ) FOR DIFFERENT MODELS USING HSI AND LIDAR DATA OF THE MG DATASET

Class No.	Traditional Learning			Convolutional Neural Networks			Transformer Models			Our Model
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	ViT	SpectralFormer	MFT	RoSENet
1	93.97	95.42±00.04	96.54	95.05±00.22	95.79±00.11	95.10±00.19	97.52±00.41	96.70±00.32	97.60±00.57	**98.76±00.09**
2	52.71	74.34±00.21	82.76	70.35±03.30	72.76±00.58	63.72±03.18	81.59±02.51	80.22±01.17	91.03±02.72	**92.63±00.75**
3	67.71	75.95±00.12	81.63	75.80±02.09	78.92±00.52	69.94±03.42	80.22±02.17	81.45±00.89	88.94±01.30	**93.17±00.84**
4	55.33	68.52±00.35	87.95	78.60±02.77	83.59±00.99	63.90±01.84	86.17±02.77	89.88±01.24	93.27±02.29	**94.93±01.08**
5	83.16	88.62±00.11	89.56	78.31±01.48	78.29±01.12	79.48±01.43	93.71±00.79	92.86±00.40	94.14±01.05	**96.15±00.36**
6	18.96	76.07±00.53	87.13	46.35±02.49	50.34±02.13	02.86±03.00	63.04±07.50	72.09±03.18	79.16±04.96	**92.87±02.17**
7	44.18	66.25±00.38	80.67	78.31±00.20	79.70±00.26	47.96±01.00	82.46±01.57	87.96±01.57	88.64±03.68	**93.43±00.23**
8	74.06	93.38±00.09	92.66	66.72±01.17	71.95±01.10	70.47±01.25	93.22±00.70	93.17±00.66	96.70±00.60	**98.13±00.23**
9	3.5	19.95±00.29	20.51	40.15±02.96	43.92±01.24	06.28±04.91	61.90±04.10	60.74±01.68	50.01±09.29	**77.28±02.14**
10	0	02.41±00.67	04.59	09.20±01.24	12.45±00.27	00.00±00.00	14.82±07.78	**27.52±05.29**	01.61±01.68	22.18±04.16
11	62.11	72.18±00.34	60.54	25.65±02.89	26.82±02.60	66.93±01.76	**75.35±06.34**	60.00±05.30	71.72±06.37	71.36±03.56
OA(%)	76.81	85.49±00.04	88.72	81.50±00.03	83.40±00.04	77.99±00.06	90.44±00.25	90.39±00.30	93.07±00.23	**95.80±00.10**
AA(%)	50.52	66.65±00.08	71.32	60.41±00.48	63.14±00.21	51.51±00.40	75.46±01.09	76.60±01.18	77.53±01.41	**84.63±00.49**
$\kappa \times 100$	68.56	80.62±00.06	84.96	75.43±00.07	77.94±00.06	70.31±00.03	87.30±00.33	87.26±00.39	90.81±00.30	**94.44±00.13**

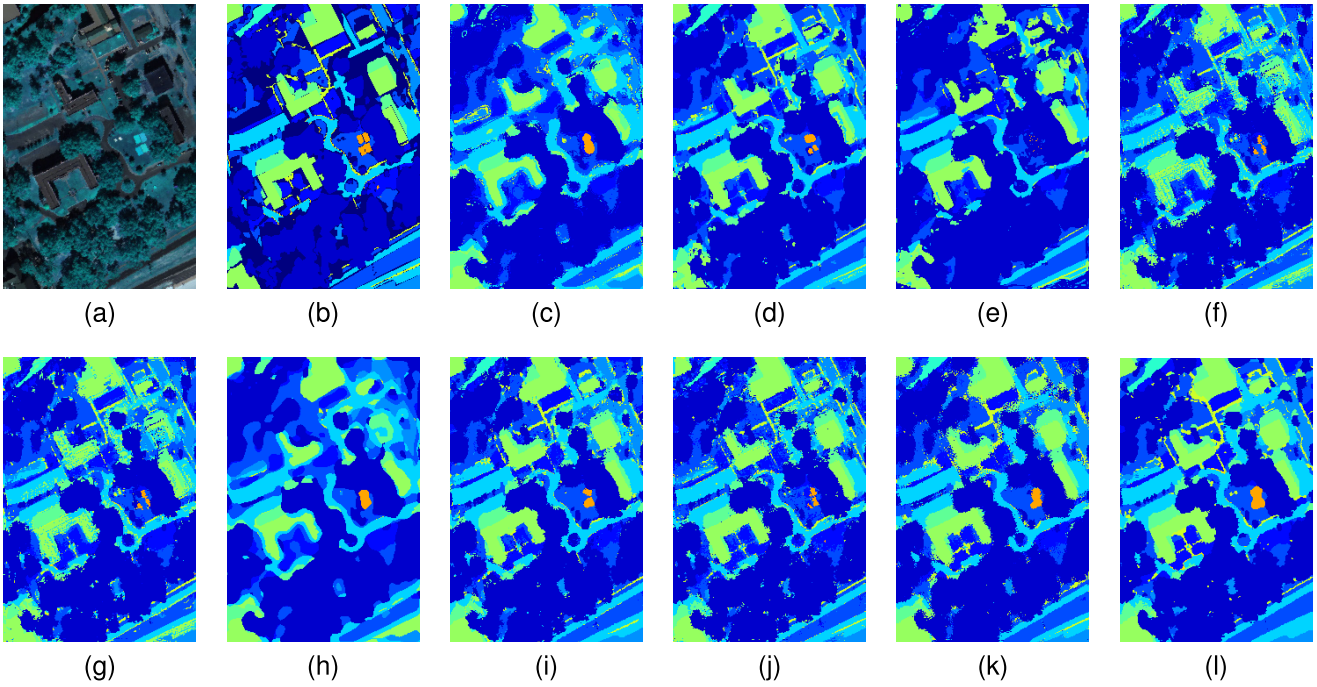


Fig. 7. Visual results of MG: (a) color composite illustration and (b) ground truth and classification maps obtained by HSI and LiDAR data for the MG dataset by (c) KNN, (d) RF, (e) SVM, (f) CNN1D, (g) CNN2D, (h) CNN3D, (i) ViT, (j) SpectralFormer, (k) MFT, and (l) RoSENet using random sampling.

coefficient, and classification accuracy for each category. In the table, bolded text represents the maximum value of the indicator in the method, while underlined text indicates the second highest value of the indicator. Evaluation results indicate that RoSENet outperforms other methods by achieving the highest OA, AA, and κ values and achieves superior classification accuracy across most categories.

Several phenomena can be observed from Table I. Among traditional classifiers, SVM demonstrates the best performance, achieving an OA of 88.72%, AA of 71.32%, and κ value of 0.8496. These values surpass other traditional classifiers, such as KNN and RF, and highlight SVM's ability to effectively process hyperspectral data due to its robust classification boundaries. However, traditional models heavily depend on spectral information and struggle to incorporate spatial data, limiting their performance on complex datasets.

For CNN models, Table I indicates that although CNN-based methods (CNN1D, CNN2D, and CNN3D) can capture local spatial features, their performance on the MG dataset does not exceed that of SVM. The best CNN model, CNN2D, attains an OA of 83.40%, an AA of 63.14%, and a κ value of 0.7794. CNNs, while effective in many other scenarios, face challenges when dealing with the fusion of hyperspectral and LiDAR data. This limitation stems from their reliance on convolutional kernels, which are adept at capturing local features but struggle to capture cross-channel spectral and spatial information.

With the introduction of transformer-based models, classification performance improves significantly. The MFT model, in particular, demonstrates superior results by effectively capturing complementary spatial and spectral information. It achieves an OA of 93.07%, an AA of 77.53%, and a κ value of 0.9081. This substantial improvement over traditional classifiers and CNNs can be attributed to transformer models' ability to capture long-range dependencies using self-attention mechanisms.

Moreover, the proposed RoSENet model further enhances performance. With its unique architectural design and

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TABLE II
COMPARISON OF OA, AA, AND KAPPA FOR DIFFERENT MODELS USING HSI AND LIDAR DATA OF THE TRENTO DATASET

Class No.	Traditional Learning			Convolutional Neural Networks			Transformer Models			Our Model
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	ViT	SpectralFormer	MFT	RoSENet
1	89.32	83.44±00.12	92.11	97.00±00.50	96.98±00.21	92.95±00.10	95.58±01.18	96.76±01.71	96.82±01.01	**98.73±00.57**
2	96.11	96.36±00.18	**99.89**	96.51±01.70	97.56±00.14	98.09±00.23	99.43±00.47	97.25±00.66	99.60±00.45	99.04±00.96
3	87.7	69.12±00.18	91.18	42.34±06.33	55.35±00.00	93.85±01.09	54.86±06.83	58.47±11.54	87.57±10.12	**98.80±01.10**
4	96.24	99.81±00.02	97.57	99.77±00.05	99.66±00.03	99.32±00.05	99.22±01.70	99.24±00.21	99.89±00.07	**100.00±00.00**
5	93.62	95.32±00.11	89.37	99.27±00.09	99.56±00.07	98.74±00.04	98.29±00.53	93.52±01.75	99.87±00.11	**99.92±00.06**
6	82.53	72.63±00.35	85.55	76.91±03.62	76.91±00.15	88.15±00.20	80.01±01.41	73.39±06.78	83.92±03.08	**94.43±01.43**
OA(%)	92.86	92.52±00.04	92.86	95.81±00.13	96.14±00.03	96.93±00.03	95.87±00.59	93.51±01.27	97.63±00.26	**99.12±00.11**
AA(%)	90.92	86.11±00.07	92.61	85.30±00.72	87.67±00.04	95.18±00.18	87.90±01.12	86.44±02.96	94.61±01.55	**98.49±00.22**
$\kappa \times 100$	90.57	90.05±00.06	90.57	94.39±00.17	94.83±00.04	95.89±00.04	94.47±00.78	91.36±01.67	96.83±00.35	**98.82±00.15**

innovative feature extraction mechanism, RoSENet not only surpasses traditional classifiers and CNNs in classification accuracy but also significantly outperforms transformer models. In the classification task on the MG dataset, RoSENet gains an OA value of 95.80%, an AA value of 84.63%, and a κ value of 0.9444. Compared to the second-best MFT model, RoSENet demonstrates an OA increase of 2.73%, an AA increase of 7.10%, and a kappa coefficient improvement of 3.63%, highlighting its stronger feature representation and ability to accurately capture similarity dependencies between feature maps. This enhances both classification accuracy and stability. As Table I shows, RoSENet consistently surpasses other models on complex datasets, and the low standard deviations across three metrics (0.10%, 0.49%, and 0.13%) further underscore its improved stability and overall superiority.

In the classification results for specific categories, our method performs best for all categories except Yellow-Curb and ClothPanels (classes 10 and 11), where it is slightly lower than SpectralFormer and ViT, respectively. Considering that these two categories have a relatively dispersed pixel distribution compared to the more concentrated distribution of other categories, along with their low contrast and somewhat blurred boundaries, Transformer-based methods generally perform better than CNN-based methods. RoSENet, which combines CNN and self-attention mechanisms, leverages the advantages of both, but in some cases, it may not achieve optimal performance for certain categories.

Next, to fully demonstrate the generalization ability of the proposed model and effectively reduce the chance factors that might arise from using a single dataset, this article carefully selects another two representative datasets—Trento and UH. These datasets not only cover different geographical regions and LC types but also incorporate non-overlapping data partitions, which significantly increase classification difficulty. This added complexity ensures that the model is trained and tested under more challenging conditions, providing a more robust verification of its generalization ability and adaptability.

The corresponding results for the Trento and UH datasets are presented in Tables II and III. The results from the Trento dataset in Table II demonstrate that CNN-based methods generally outperform traditional methods due to the relatively low noise in the Trento dataset, where CNN-based methods are able to fully leverage their powerful spatial feature extraction capabilities. However, they still do not perform as well as the MFT model. It is clear that the proposed model (see Table II) significantly outperforms other models. On the Trento dataset, which includes both HSI and LiDAR data, the proposed model gets improvements of 1.49%, 3.88%, and 1.99% in OA, AA, and κ values, respectively, compared to the second-best model, MFT. On the Trento dataset, RoSENet achieves the best results for all classes except Buildings (class 2), where the SVM performs the best. Our RoSENet model also performs well on class 2, with the classification accuracy beyond 99%.

On the UH dataset (see Table III), although the noise level is not as high as in the MG dataset, the non-overlapping data partitioning introduces significant classification difficulty. This makes the UH dataset challenging for most models. CNN-based methods generally perform worse on this dataset, with the best CNN model (CNN2D) obtaining an OA of 71.88%, an AA of 70.51%, and a κ value of 0.6956. In contrast, transformer models demonstrate a significant advantage due to their ability to extract global features effectively. The proposed model improves OA, AA, and κ values by 1.38%, 1.77%, and 1.50%, respectively, compared to MFT, reaching an OA of 88.96%, an AA of 90.78%, and a κ value of 0.8802. However, for the Grass-stressed, Tree, and Commercial classes (i.e., classes 2, 4, and 8), our method does not surpass MFT. This is mainly because Transformer-based models excel at capturing global dependencies, which is particularly beneficial for objects with clear shape structures and for handling large-scale texture and color variations. In contrast, these three classes exhibit more irregular textures and less distinct structural patterns, making them less suited for our model's strengths. Nevertheless, our RoSENet model still delivers competitive performance on these classes.

In brief, through the quantitative evaluation in the above experiments, RoSENet has demonstrated the best performance among all the compared models across different datasets and various scenarios (i.e., random sampling and non-overlapping sampling), indicating its effectiveness and generalization ability.

D. Visual Comparison

Next, a qualitative evaluation is performed by visualizing the classification maps derived from different methods, as shown in Figs. 7–9, which illustrate results obtained from multimodal datasets (MG, Trento, UH). Although traditional classifiers such as KNN, RF, and SVM produce

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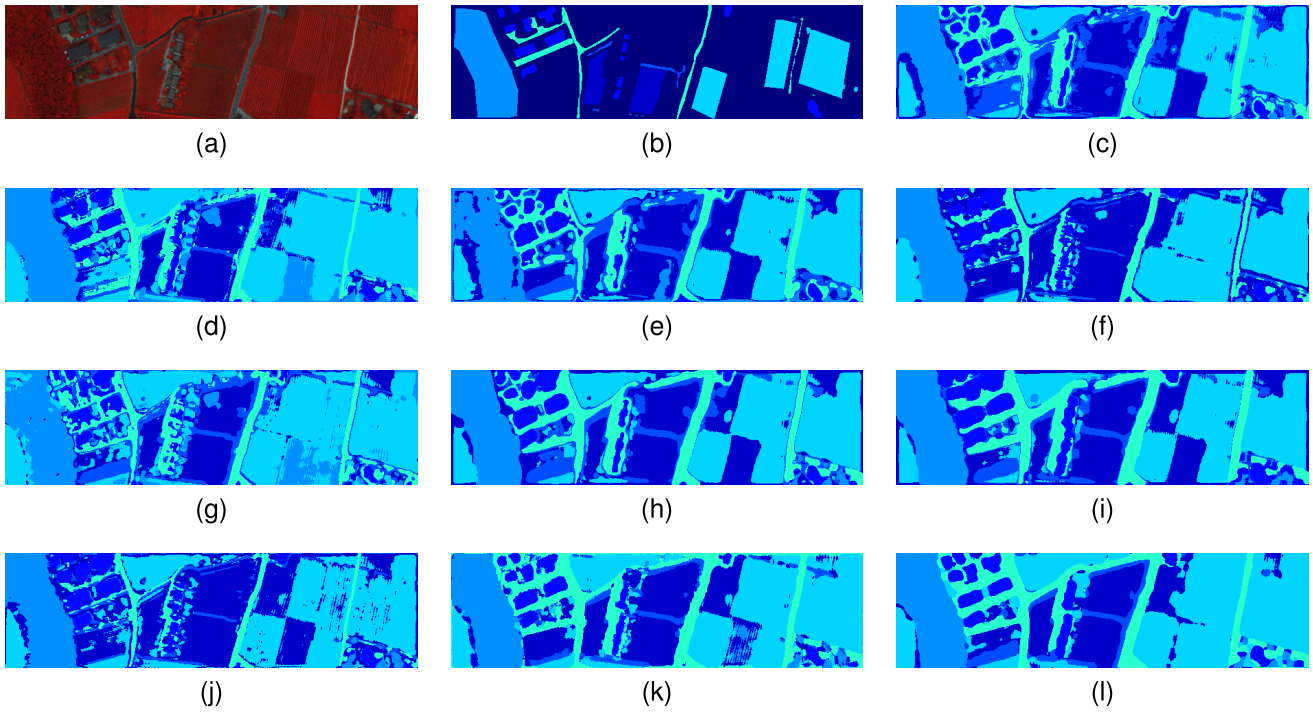


Fig. 8. Visual results of Trento: (a) color composite illustration and (b) ground truth and classification maps obtained by HSI and LiDAR data for the Trento dataset by (c) KNN, (d) RF, (e) SVM, (f) CNN1D, (g) CNN2D, (h) CNN3D, (i) ViT, (j) SpectralFormer, (k) MFT, and (l) RoSENet using non-overlapping training/test samples.

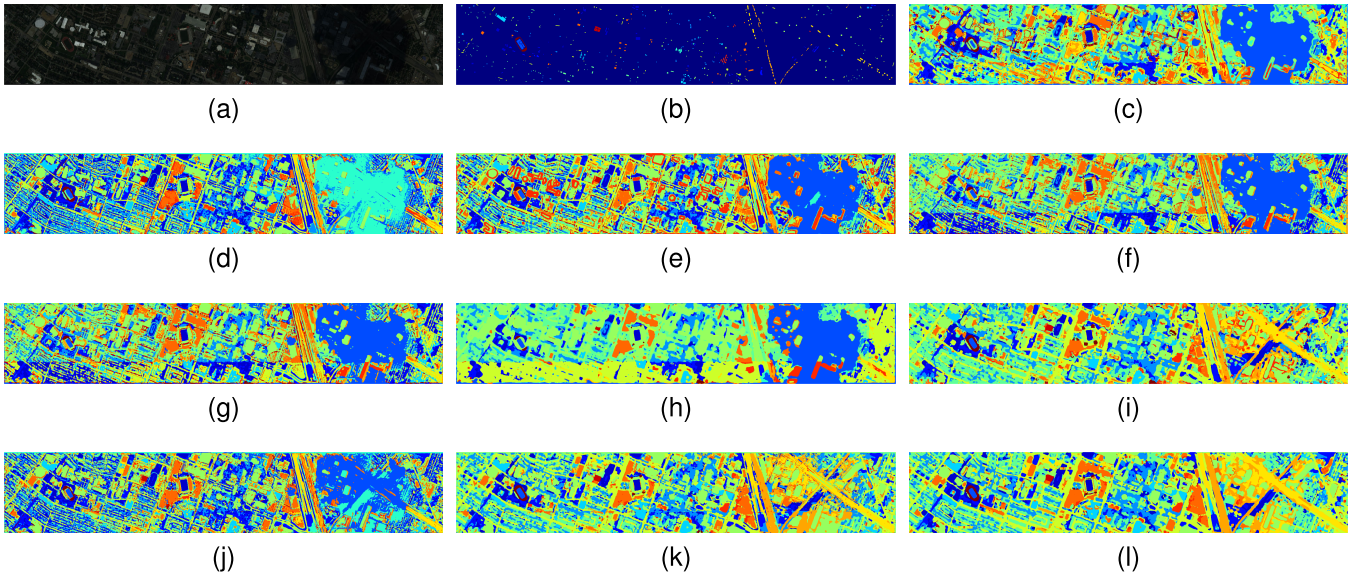


Fig. 9. Visual results of UH: (a) color composite illustration and (b) ground truth and classification maps obtained for the UH dataset by (c) KNN, (d) RF, (e) SVM, (f) CNN1D, (g) CNN2D, (h) CNN3D, (i) ViT, (j) SpectralFormer, (k) MFT, and (l) RoSENet using non-overlapping training/test samples.

rich classification maps, these maps still contain noise in the boundary areas of LC categories due to their reliance solely on spectral information from the HSI data [Figs. 7(c)–(e), 8(c)–(e), and 9(c)–(e)]. In contrast, deep learning models, with their trainable feature mining capabilities, are successful in capturing the nonlinear connections between the input and output feature maps. As a result, CNNs generate comparatively uniform classification maps with clearly distinguishable boundary areas of LC categories [Figs. 7(f)–(h), 8(f)–(h), and 9(f)–(h)]. Transformer models perform remarkably well in HSI classification because they can fully extract high-level sequential abstractions from HSI data. Consequently, the visual quality of the classification maps is superior to that of classical networks [Figs. 7(i)–(k), 8(i)–(k), and 9(i)–(k)].

Through the more effective enhancement of neighboring spectral information and the transfer of positional information between the layers, the proposed RoSENet can generate highly preferable classification maps with clear texture and edge details, outperforming ViT, SpectralFormer, and MFT. As seen in the generated classification maps [Figs. 7(l), 8(l), and 9(l), considering information from other modalities during feature extraction and learning resulted in significant improvements in class boundaries and visual quality. Due to the fully exploited relationships between the central spectral vector and those

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TABLE III
COMPARISON OF OA, AA, AND KAPPA FOR DIFFERENT MODELS USING HSI AND LIDAR DATA OF THE UH DATASET

Class No.	Traditional Learning			Convolutional Neural Networks			Transformer Models			Our Model
	KNN	RF	SVM	CNN1D	CNN2D	CNN3D	ViT	SpectralFormer	MFT	RoSENet
1	77.59	78.84±0.07	80.06	81.32±0.16	82.62±0.08	82.30±0.29	82.33±0.69	82.40±0.32	81.99±3.41	**83.01±0.07**
2	80.73	72.90±0.43	80.17	81.92±0.19	81.83±0.66	78.70±0.98	78.70±2.19	82.82±1.79	**86.52±3.73**	85.78±3.24
3	98.22	98.18±0.08	100.00	57.49±0.61	62.90±0.37	96.96±0.47	76.95±4.08	64.61±4.57	98.61±0.61	**99.42±0.52**
4	81.16	78.32±0.17	80.21	86.74±0.23	88.57±0.04	80.49±1.57	91.15±2.47	94.44±3.14	**95.32±1.52**	94.01±3.11
5	92.23	89.69±0.21	97.82	95.93±0.27	97.19±0.19	98.11±0.54	97.46±1.55	97.64±0.60	99.38±0.27	**100.00±0.00**
6	76.92	94.13±0.56	95.80	58.04±0.57	65.03±1.98	73.89±1.84	85.87±6.15	88.46±1.88	94.20±6.49	**96.29±4.45**
7	81.34	78.97±0.46	80.60	77.95±0.88	80.22±1.42	81.09±0.75	85.60±1.76	87.23±1.16	86.20±2.45	**89.20±1.83**
8	43.87	52.01±1.24	56.70	54.42±2.33	60.81±4.32	44.63±6.52	75.43±5.02	72.72±3.26	**82.36±5.78**	80.61±1.98
9	70.54	76.84±0.13	80.36	66.13±0.19	67.74±1.08	74.76±3.84	79.58±2.67	82.41±2.40	84.24±7.06	**87.95±1.81**
10	43.92	38.42±0.28	**67.28**	47.30±2.18	51.74±1.33	37.52±11.23	45.82±2.53	45.35±2.66	59.24±4.50	62.68±5.84
11	64.90	65.37±0.56	80.36	44.40±2.06	39.91±3.52	40.80±8.50	67.07±3.05	63.79±3.32	97.25±1.64	**97.93±2.56**
12	82.13	82.09±0.18	90.78	63.66±8.15	82.20±3.85	66.38±2.36	81.67±3.18	82.33±1.57	90.80±1.65	**93.97±2.20**
13	60.35	67.82±0.59	82.46	50.41±1.86	52.40±1.29	68.77±12.16	59.68±6.94	57.12±4.62	82.11±4.85	**90.95±2.15**
14	81.38	82.27±0.40	89.88	86.50±0.69	92.44±2.89	92.85±1.16	88.17±4.46	89.19±4.51	98.87±2.26	**100.00±0.00**
15	89.43	95.64±0.36	97.46	40.10±3.20	52.08±4.14	96.90±1.41	75.98±1.35	74.60±4.32	98.12±2.26	**99.80±0.50**
OA(%)	73.66	73.86±0.13	81.47	68.11±0.29	71.88±0.36	71.41±0.35	78.22±0.72	78.20±0.80	87.58±0.57	**88.96±0.26**
AA(%)	74.98	76.77±0.12	84.00	66.15±0.09	70.51±0.27	74.28±0.30	78.10±1.11	77.67±0.69	89.01±0.56	**90.78±0.40**
$\kappa \times 100$	71.57	71.81±0.13	79.99	65.48±0.31	69.56±0.39	69.12±0.36	76.41±0.77	76.43±0.86	86.52±0.62	**88.02±0.29**

TABLE IV
OA, AA, AND KAPPA VALUES OF ROSENET UNDER THE SCENARIOS OF USING SINGLE-MODAL HSI DATA AND MULTIMODAL HSI AND LIDAR DATA

Metric	HSI			HSI + LiDAR		
	MG	TRENTO	UH	MG	TRENTO	UH
OA(%)	95.60 ± 00.23	97.48 ± 00.25	87.64 ± 00.39	**95.80±00.10**	**99.12±00.11**	**88.96±00.26**
AA(%)	84.17 ± 01.02	95.92 ± 00.41	89.86 ± 00.38	**84.63±00.49**	**98.49±00.22**	**90.78±00.40**
$\kappa \times 100$	94.17 ± 00.31	96.63 ± 00.34	86.62 ± 00.41	**94.44±00.13**	**98.82±00.15**	**88.02±00.29**

situated in the surrounding area, as well as the thorough extraction of spatial information without losing spectral information, the proposed RoSENet demonstrates more noticeable improvements. The extraction of these relationships helps to find robust nonlinear sequential relationships from HSI and LiDAR data, significantly improving the efficiency of complementary information extraction. Consequently, it is possible to generate classification maps that contain a higher degree of reproduction and more realistic and precise details in LC areas.

E. Effectiveness of Multimodal Data

In addition, we conduct experiments comparing the performance of single-modal data (HSI only) and multimodal data (HSI and LiDAR). As shown in Table IV, in this set of experiments, the proposed RoSENet on the MG dataset, the improvements in OA, AA, and kappa values are 0.20%, 0.46%, and 0.27%, respectively, compared to using single-modal HSI data. On the Trento dataset, the improvements in OA, AA, and kappa values are 1.64%, 2.57%, and 2.19%, respectively, due to using multimodal data. Similarly, the UH dataset containing HSI and LiDAR data achieves improvements in OA, AA, and kappa values of 1.32%, 0.92%, and 1.40%, respectively. On the whole, the multimodal classification that includes LiDAR data performs better than the single-modal classification using only HSI data across all three datasets. Especially for the Trento and UH datasets with more complex non-overlapping sampling, such an advantage becomes more significant, verifying the necessity of using multimodal data.

F. Ablation Experiments

The ablation study aims to explore the specific contributions of three core modules in RoSENet: RFE, SASSC, and ESSCA by comparing the full model with its variants across the MG, Trento, and UH datasets. Table V clearly outlines how each module affects the classification accuracy, with OA, AA, and κ values used as key evaluation metrics.

1) *Full model (RFE + SASSC + ESSCA):* The highest level of efficiency across all datasets is achieved by combining all three modules, as shown by the OA, AA, and κ scores. This proves that RoSENet effectively captures complementary spatial and spectral information through the integration of these modules. For example, on the MG dataset, the full model achieves an OA of 95.80%, AA of 84.63%, and κ of 0.9444, reflecting its overall effectiveness.
2) *Without RFE (SASSC + ESSCA):* By removing RFE, we observe a slight drop in performance on MG and Trento datasets and a notable drop on the UH dataset. This indicates that RFE's rotation-based augmentation holds a dominant position in the promotion of the diversity of input data by supplying subtle information of different directions through small-angle rotation, especially for complex datasets like UH, where it helps mitigate the challenge posed by non-overlapping data partitions. On the UH dataset, the absence of RFE results in a reduction in OA by 1.04%, which is a noticeable decrease compared to the full model. Comparing three datasets by observing Figs. 4–6, we can find that MG and Trento datasets are mainly composed of large objects, while the UH dataset contains many scattered small objectives with different directions. Without the help of RFE, the model with only SASSC and ESSCA can still obtain good results on the MG and Trento datasets, just depending on the strong spectral–spatial feature extraction ability of SASSC and ESSCA. However, the performance of the model without RFE degrades obviously on the UH dataset, indicating

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TABLE V

OA, AA, AND KAPPA VALUES OBTAINED BY DIFFERENT VARIANTS OF ROSENET USING DIFFERENT MODULE (RFE, SASSC, ESSCA) COMBINATIONS ON THE MG, TRENTO, AND UH DATASETS

DataSet	RFE	SASSC	ESSCA	OA (%)	AA (%)	$\kappa\times$ 100
MG	✓	✓	✓	**95.80 ± 00.10**	**84.63 ± 00.49**	**94.44 ± 00.13**
	✗	✓	✓	95.60 ± 00.26	84.31 ± 00.70	94.18 ± 00.35
	✓	✗	✓	93.01 ± 00.19	77.96 ± 00.53	90.73 ± 00.25
	✓	✓	✗	90.88 ± 00.11	74.71 ± 00.33	87.91 ± 00.15
TRENTO	✓	✓	✓	**99.12 ± 00.11**	**98.49 ± 00.22**	**98.82 ± 00.15**
	✗	✓	✓	98.99 ± 00.14	98.12 ± 00.26	98.65 ± 00.19
	✓	✗	✓	97.89 ± 00.50	97.21 ± 00.50	97.18 ± 00.67
	✓	✓	✗	98.69 ± 00.11	97.99 ± 00.16	98.25 ± 00.14
UH	✓	✓	✓	**88.96 ± 00.26**	**90.78 ± 00.40**	**88.02 ± 00.29**
	✗	✓	✓	87.92 ± 00.54	89.89 ± 00.42	86.93 ± 00.59
	✓	✗	✓	87.01 ± 00.46	88.66 ± 00.34	85.91 ± 00.50
	✓	✓	✗	87.45 ± 00.35	89.30 ± 00.29	86.41 ± 00.38

that RFE is crucial for the model to capture the subtle direction information.

3) *Without SASSC (RFE + ESSCA):* Excluding SASSC results in a sharper performance decline, particularly on the Trento and UH datasets, indicating that SASSC is especially beneficial to the difficult scenarios with non-overlapping sampling, because SASSC is crucial for preserving the critical spectral features and suppressing the redundant spectral information while extracting spatial features by using self-similarity attention maps. For instance, on the UH dataset, the OA drops by 1.95%, and the AA drops significantly by 2.12%. This emphasizes SASSC's role in effectively fusing spectral–spatial information and suppressing the redundant spectral information, making it an indispensable component of the model.

4) *Without ESSCA (RFE + SASSC):* Removing ESSCA leads to an obvious reduction in three metrics, especially on the MG dataset. Such a degradation lies in that ESSCA contributes to the global feature representation and enhances the model's ability to capture long-range dependencies between spectral and spatial data, allowing the model to automatically focus on important regions and suppress the noise in the datasets. On the MG dataset, which has strong noise, OA values decline by 4.92%, and AA values fall by 9.92%, demonstrating the importance of the global feature attention mechanism provided by ESSCA as well as its generalization, especially in noisy environments.

The ablation study confirms that each module in RoSENet plays a distinct role in improving classification accuracy, with the combined use of all three providing the highest results across different datasets. This demonstrates the model's adaptability and superior performance in handling multimodal data.

G. Hyperparameter Analysis, Complexity Evaluation, and Further Discussions

1) Analysis for Different Patch Sizes: Here, we use a fixed training ratio of 5% and randomly select training samples for all three datasets with patch sizes of 3×3, 5×5, 7×7, 9×9, 11×11, and 13×13 to conduct experiments (see Table VI). From Table VI, we can see that the proposed RoSENet shows highly stable performance,

TABLE VI

OA, AA, AND KAPPA VALUES OF ROSENET ON THE MG (HSI + LIDAR), TRENTO (HSI + LIDAR), AND UH (HSI + LIDAR) DATASETS WHEN RANDOMLY SELECTING SAMPLES WITH DIFFERENT PATCH SIZES UNDER A FIXED TRAINING RATIO OF 5%

Patch Size	Metric	MG	TRENTO	UH
3×3	OA(%)	95.98 ± 00.07	99.85 ± 00.01	98.47 ± 00.16
	AA(%)	86.85 ± 00.44	99.40 ± 00.04	98.47 ± 00.15
	$\kappa\times$ 100	94.68 ± 00.10	99.81 ± 00.01	98.35 ± 00.18
5×5	OA(%)	95.93 ± 00.13	99.86 ± 00.01	98.53 ± 00.10
	AA(%)	86.55 ± 00.71	99.41 ± 00.05	98.56 ± 00.11
	$\kappa\times$ 100	94.62 ± 00.16	99.81 ± 00.02	98.41 ± 00.11
7×7	OA(%)	96.03 ± 00.20	99.86 ± 00.03	98.46 ± 00.14
	AA(%)	87.03 ± 01.39	99.44 ± 00.07	98.49 ± 00.13
	$\kappa\times$ 100	94.75 ± 00.27	99.81 ± 00.04	98.34 ± 00.15
9×9	OA(%)	96.02 ± 00.10	99.87 ± 00.02	**98.60 ± 00.12**
	AA(%)	86.88 ± 00.47	99.43 ± 00.04	**98.59 ± 00.13**
	$\kappa\times$ 100	94.74 ± 00.14	99.83 ± 00.03	**98.48 ± 00.13**
11×11	OA(%)	96.05 ± 00.15	99.86 ± 00.02	98.48 ± 00.16
	AA(%)	86.96 ± 00.67	99.42 ± 00.06	98.47 ± 00.25
	$\kappa\times$ 100	94.77 ± 00.20	99.81 ± 00.03	98.35 ± 00.18
13×13	OA(%)	**96.05 ± 00.19**	**99.88 ± 00.02**	98.49 ± 00.12
	AA(%)	**87.20 ± 01.13**	**99.48 ± 00.05**	98.53 ± 00.09
	$\kappa\times$ 100	**94.77 ± 00.25**	**99.84 ± 00.03**	98.37 ± 00.13

with minimal differences in performance across different patch sizes. Since the classification difficulty of random sampling is lower than non-overlapping samples, the accuracy of random sampling on the UH and Trento datasets in this part improves significantly compared to the previous non-overlapping sampling. As shown in Table VI, across all the datasets evaluated in the experiments, the proposed model achieves the best and most stable AA, OA, and kappa values on the UH dataset using a 9×9 window. On the Trento and MG datasets, the model performs best when using a 13×13 window. From Table VI, it can be observed that, although the AA, OA, and Kappa values are not the highest when using a 9×9 window on the MG and Trento, they still achieve consistently strong results across all three metrics, indicating an overall favorable performance. However, as the patch size decreased, the performance on the UH and Trento datasets was only slightly affected, while the performance on the MG dataset declined. It should be noted that expanding the spatial window size leads to increased redundancy. Therefore, to strike a balance and ensure consistent performance across the three datasets, we used patches of 11×11 in the entire evaluation.

2) Parameter and Time Complexity Evaluation: We compare the proposed RoSENet with other transformer networks in terms of OA, number of parameters, and time complexity, as shown in Fig. 10(a)–(f). They, respectively, show the change of OA with a number of parameters and training time for the UH, Trento, and MG datasets. Fig. 10(a)–(c) reflects the relationship between the number of parameters and OA for transformer-based models and RoSENet, while Fig. 10(d)–(f) reflects the relationship between training time and OA for these methods across the corresponding datasets. On the MG dataset, the proposed RoSENet demonstrates clear efficiency, using fewer parameters than the second-best model, MFT, In the Trento dataset, RoSENet also performs well in terms of parameter count, significantly lower than that of MFT, and its parameter count and training time are even comparable to earlier models. However, RoSENet's parameter count and

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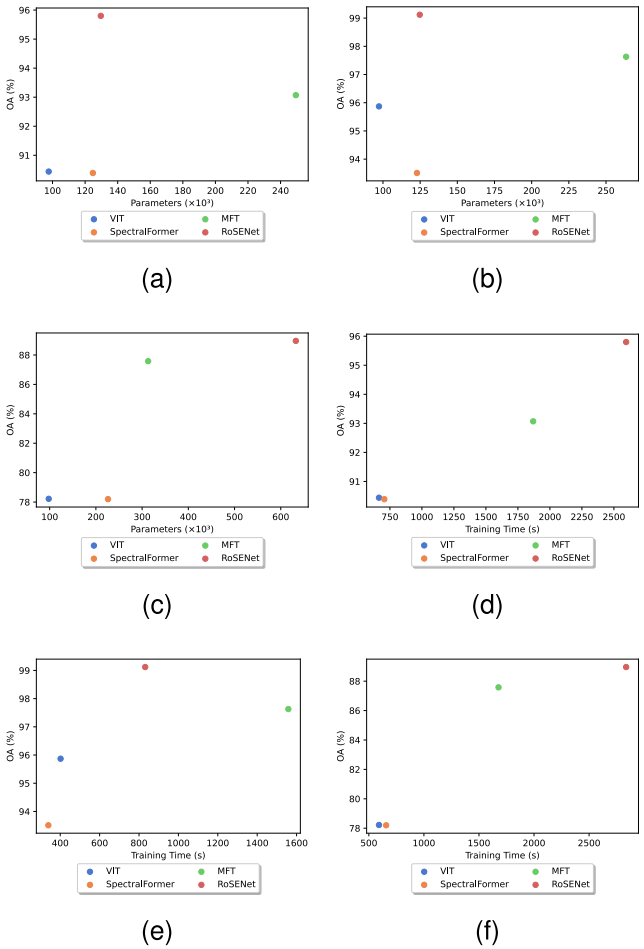


Fig. 10. Comparison of OA, number of parameters, and training time of the RoSENet model on the MG, Trento, and UH datasets. (a) MG (number of parameters). (b) Trento (number of parameters). (c) UH (number of parameters). (d) MG (time). (e) Trento (time). (f) UH (time).

training time are higher than those of other models in the UH dataset, and RoSENet has the longest training time in the MG dataset. Considering the improvement in classification accuracy and model stability, such extra computational cost is reasonable. Just as observed in the MG dataset, the proposed RoSENet acquires the improved OA by 2.73% and AA by 7.10% compared to the second-best model, MFT. The performance and complexity of the model should be considered by weighing a substantial increase in classification effectiveness against a limited additional computational cost. In short, RoSENet is both effective and efficient in terms of classification accuracy and computational complexity.

3) Comparison Experiments Between RoSENet and Its Variants Using Different Data Augmentation Methods: Common data augmentation techniques typically involve rotation, flipping, etc., with rotation often being performed at 90° intervals. We design the augmentation strategy by only using a multiangle rotation with a 45° interval for the RFE module. Its advantage lies in that it allows the model to extract more detailed directional semantic information for the subsequent SASSC and ESSCA modules while introducing minimal additional computational cost. To demonstrate the advantages of RFE, we have conducted an experiment

TABLE VII

OA, AA, AND KAPPA VALUES OBTAINED BY DIFFERENT VARIANTS OF ROSENET USING DIFFERENT DATA AUGMENTATION MODULES ON THE UH DATASET

Metric	UH				
	R90°	R90°-lr	R90°-ud	R90°-lrud	RFE
OA(%)	88.30 ± 0.30	88.33 ± 0.29	88.09 ± 0.44	88.02 ± 0.26	**88.96 ± 0.27**
AA(%)	90.20 ± 0.32	90.20 ± 0.29	90.11 ± 0.39	89.98 ± 0.30	**90.78 ± 0.40**
$\kappa\times$ 100	87.32 ± 0.33	87.35 ± 0.33	87.11 ± 0.47	87.03 ± 0.28	**88.03 ± 0.29**

TABLE VIII

OA, AA, AND KAPPA VALUES FOR DIFFERENT MODELS USING HSI AND MULTISPECTRAL DATA OF THE UH DATASET

Metric	UH			
	SVM	3D-CNN	MFT	RoSENet
OA(%)	94.22	81.62 ± 2.12	96.66 ± 0.45	**97.53 ± 0.20**
AA(%)	93.78	80.71 ± 2.59	96.26 ± 0.58	**97.09 ± 0.18**
$\kappa\times$ 100	93.75	80.11 ± 2.30	96.39 ± 0.49	**97.33 ± 0.22**

comparing RoSENet with its variants on the UH dataset. The results are shown in Table VII, where R90° represents a basic variant of RoSENet that uses [0°, 90°, 180°, 270°] as the rotation angles for data augmentation. The suffix -lr, -ud, and -lrud represent the addition of horizontal flipping, vertical flipping, and both of them to the variant with 90° intervals, respectively.

Table VII shows that our proposed method with a 45° interval enables RFE to perform better than four traditional methods. Among traditional methods, R90°-lr performs similar to the basic variant R90°, while R90°-ud and R90°-lrud perform much worse. This indicates that, at least in our model, flipping operations do not necessarily further improve performance and may even lead to degradation. In contrast, our augmentation strategy of only using rotations with a small-angle rotation interval is more effective because RFE can cooperate with the subsequent similarity attention well by providing more detailed directional semantic information.

4) Comparison Experiments Using HSI and MS Data: Considering that the collection of LiDAR data is costly and perhaps inaccessible in some circumstances, we compare RoSENet with other methods on the UH dataset, using both hyperspectral and multispectral data for classification. The experimental results are shown in Table VIII, where the comparison methods are representative models (SVM/3DCNN/MFT) selected from various approaches. The patch size is 11, and 5% of the samples are randomly selected from each class as training samples.

As seen in Table VIII, RoSENet outperforms other methods on all three metrics. This indicates that RoSENet is also applicable to the fusion of other modalities with HSIs and performs excellently. Furthermore, RoSENet not only achieves the best mean values but also the lowest standard deviation, demonstrating its high stability.

5) Comparison Experiments With Fewer Training Samples: To verify the effectiveness of RoSENet with fewer training samples, we randomly sample 0.5% samples per class as the training samples and select representative methods (SVM/3DCNN/MFT) from various approaches as comparison

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TABLE IX
OA, AA, AND KAPPA VALUES OF ROSENET ON THE MG (HSI + LIDAR), TRENTO (HSI + LIDAR), AND UH (HSI + LIDAR) DATASETS WHEN RANDOMLY SELECTING 0.5% SAMPLES PER CLASS

Metric	MG			
	SVM	3D-CNN	MFT	RoSENet
OA(%)	81.75	78.15 ± 0.86	83.62 ± 0.63	**86.75 ± 0.41**
AA(%)	59.78	48.01 ± 1.38	60.20 ± 1.63	**62.13 ± 0.89**
$\kappa\times$ 100	75.74	70.90 ± 1.20	78.25 ± 0.87	**82.37 ± 0.53**

Metric	Trento			
	SVM	3D-CNN	MFT	RoSENet
OA(%)	97.27	94.76 ± 0.24	98.36 ± 0.24	**99.52 ± 0.11**
AA(%)	93.55	84.12 ± 0.56	95.70 ± 1.02	**98.44 ± 0.54**
$\kappa\times$ 100	96.35	92.98 ± 0.32	97.81 ± 0.33	**99.36 ± 0.15**

Metric	UH			
	SVM	3D-CNN	MFT	RoSENet
OA(%)	73.55	40.77 ± 2.18	73.94 ± 2.48	**82.80 ± 0.48**
AA(%)	72.36	35.90 ± 2.33	74.06 ± 2.85	**84.28 ± 0.37**
$\kappa\times$ 100	71.40	35.61 ± 2.38	71.81 ± 2.68	**81.40 ± 0.52**

methods. We test them on three benchmark datasets, and the results are shown in Table IX.

Comparing the results in Tables VI and IX, we can observe that when the proportion of randomly sampled training samples per class is reduced from 5% to 0.5%, the classification accuracy of RoSENet decreases across all three datasets. However, compared to the other three methods, RoSENet still shows significant advantages and can achieve satisfactory classification results even with fewer training samples. Among the three comparison methods, the traditional deep model 3D-CNN performs the worst, with its classification accuracy even lower than that of the traditional SVM classifier. This is because traditional deep models heavily rely on a large number of labeled samples for training, and the performance of 3D-CNN just aligns with such a conventional expectation. Moreover, both 3D-CNN and SVM are single-modal methods that do not consider how to enhance classification performance through effective fusion of different modalities. As multimodal methods, RoSENet and MFT achieve relatively satisfactory classification results through effective multimodal data fusion strategies. Furthermore, compared to MFT, RoSENet still demonstrates a significant advantage, especially on the UH dataset, where RoSENet outperforms MFT by 8.86% in OA, 10.22% in AA, and 9.59% in the kappa coefficient. On each dataset, RoSENet maintains very stable performance, especially in maintaining AA values, significantly outperforming other methods. By combining the self-similarity attention mechanism of convolution for fully mining the data semantic information, RoSENet fuses HSI and LiDAR data, effectively utilizing the complementary nature of these two modalities and alleviating the need for large-scale labeled data to some extent.

V. CONCLUSION

This article proposes a novel multimodal fusion method based on the RoSENet to address the LC classification problem by using HSI and LiDAR data. Compared to other methods, RoSENet captures and fuses the spectral and spatial information from multimodal data in a more effective manner by introducing the SASSC module and the ESSCA module, significantly improving classification accuracy. Additionally, the use of the RFE module enhances the model's robustness to directional changes in the data, improving its generalization ability. Experimental results show that RoSENet demonstrates excellent performance across three multimodal datasets, surpassing traditional machine learning methods, classical CNN models, and the state-of-the-art Transformer models.

Future work could focus on enhancing the computational efficiency of RoSENet further, making it more practical for larger-scale datasets. Moreover, exploring the integration of other types of RS data (such as multispectral images and radar data) into the model could broaden its applicability in broader application scenarios.

REFERENCES

[1] E. Bartholomé and A. S. Belward, "GLC2000: A new approach to global land cover mapping from Earth observation data," *Int. J. Remote Sens.*, vol. 26, no. 9, pp. 1959–1977, May 2005.

[2] M. Ahmad et al., "Hyperspectral image classification—Traditional to deep models: A survey for future prospects," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 968–999, 2022.

[3] F. Ullah, I. Ullah, R. U. Khan, S. Khan, K. Khan, and G. Pau, "Conventional to deep ensemble methods for hyperspectral image classification: A comprehensive survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 3878–3916, 2024.

[4] M. Zhang, W. Li, X. Zhao, H. Liu, R. Tao, and Q. Du, "Morphological transformation and spatial-logical aggregation for tree species classification using hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501212.

[5] J. Jang, S. Oh, Y. Kim, D. Seo, Y. Choi, and H. J. Yang, "M^{2}SODAI: Multi-modal maritime object detection dataset with RGB and hyperspectral image sensors," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 36, A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, Eds., Red Hook, NY, USA: Curran Associates, 2023, pp. 53831–53843.

[6] X. Wu, D. Hong, and J. Chanussot, "UIU-Net: U-Net in U-Net for infrared small object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 364–376, 2023.

[7] Y. Ma, S. Chen, S. Ermon, and D. B. Lobell, "Transfer learning in environmental remote sensing," *Remote Sens. Environ.*, vol. 301, Feb. 2024, Art. no. 113924.

[8] H. Fang et al., "Scene change detection by differential aggregation network and class probability-based fusion strategy," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5406918.

[9] D. Hong et al., "Interpretable hyperspectral artificial intelligence: When nonconvex modeling meets hyperspectral remote sensing," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 2, pp. 52–87, Jun. 2021.

[10] D. Hong, N. Yokoya, J. Chanussot, and X. X. Zhu, "An augmented linear mixing model to address spectral variability for hyperspectral unmixing," *IEEE Trans. Image Process.*, vol. 28, no. 4, pp. 1923–1938, Apr. 2019.

[11] P. Ghamisi, J. A. Benediktsson, and S. Phinn, "Land-cover classification using both hyperspectral and LiDAR data," *Int. J. Image Data Fusion*, vol. 6, no. 3, pp. 189–215, Jul. 2015.

[12] U. Heiden, W. Heldens, S. Roessner, K. Segl, T. Esch, and A. Mueller, "Urban structure type characterization using hyperspectral remote sensing and height information," *Landscape Urban Planning*, vol. 105, no. 4, pp. 361–375, Apr. 2012.

[13] S. K. Roy, S. Manna, T. Song, and L. Bruzzone, "Attention-based adaptive spectral–spatial kernel ResNet for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7831–7843, Sep. 2021.

[14] S. K. Roy, R. Mondal, M. E. Paoletti, J. M. Haut, and A. Plaza, "Morphological convolutional neural networks for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 8689–8702, 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:57:43 UTC from IEEE Xplore. Restrictions apply.

[15] D. Hong et al., "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5518615.

[16] S. K. Roy, G. Krishna, S. R. Dubey, and B. B. Chaudhuri, "HybridSN: Exploring 3-D–2-D CNN feature hierarchy for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 277–281, Feb. 2020.

[17] M. Ahmad, A. M. Khan, M. Mazzara, S. Distefano, M. Ali, and M. S. Sarfraz, "A fast and compact 3-D CNN for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[18] D. Hong, J. Yao, D. Meng, Z. Xu, and J. Chanussot, "Multimodal GANs: Toward crossmodal hyperspectral–multispectral image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5103–5113, Jun. 2021.

[19] D. Hong, N. Yokoya, J. Chanussot, J. Xu, and X. X. Zhu, "Joint and progressive subspace analysis (JPSA) with spatial–spectral manifold alignment for semisupervised hyperspectral dimensionality reduction," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3602–3615, Jul. 2021.

[20] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[21] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.

[22] X. Wang, Y. Feng, R. Song, Z. Mu, and C. Song, "Multi-attentive hierarchical dense fusion net for fusion classification of hyperspectral and LiDAR data," *Inf. Fusion*, vol. 82, pp. 1–18, Jun. 2022.

[23] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022, doi: 10.1109/LGRS.2020.3017414.

[24] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[25] K. Makantasis, K. Karantzalos, A. Doulamis, and N. Doulamis, "Deep supervised learning for hyperspectral data classification through convolutional neural networks," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2015, pp. 4959–4962.

[26] A. B. Hamida, A. Benoit, P. Lambert, and C. B. Amar, "3-D deep learning approach for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 8, pp. 4420–4434, Aug. 2018.

[27] C. Mu, Y. Liu, X. Yan, A. Ali, and Y. Liu, "Few-shot open-set hyperspectral image classification with adaptive threshold using self-supervised multitask learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5526618.

[28] W. Li, C. Chen, M. Zhang, H. Li, and Q. Du, "Data augmentation for hyperspectral image classification with deep CNN," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 4, pp. 593–597, Apr. 2019.

[29] Q. Hao, S. Li, and X. Kang, "Multilabel sample augmentation-based hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 6, pp. 4263–4278, Jun. 2020.

[30] W. Wang, Y. Chen, X. He, and Z. Li, "Soft augmentation-based Siamese CNN for hyperspectral image classification with limited training samples," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[31] X. Zheng, H. Sun, X. Lu, and W. Xie, "Rotation-invariant attention network for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 31, pp. 4251–4265, 2022.

[32] L. Chen, Y. Wei, Z. Yao, E. Chen, and X. Zhang, "Data augmentation in prototypical networks for forest tree species classification using airborne hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4410116.

[33] M. Khodadadzadeh, J. Li, S. Prasad, and A. Plaza, "Fusion of hyperspectral and LiDAR remote sensing data using multiple feature learning," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2971–2983, Jun. 2015.

[34] M. Dalla Mura, J. A. Benediktsson, B. Waske, and L. Bruzzone, "Morphological attribute profiles for the analysis of very high resolution images," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 10, pp. 3747–3762, Oct. 2010.

[35] W. Liao, R. Bellens, A. Pizurica, S. Gautama, and W. Philips, "Graph-based feature fusion of hyperspectral and LiDAR remote sensing data using morphological features," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, vol. 7, Jul. 2013, pp. 4942–4945.

[36] B. Rasti, P. Ghamisi, and R. Gloaguen, "Hyperspectral and LiDAR fusion using extinction profiles and total variation component analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3997–4007, Jul. 2017.

[37] A. Merentitis, C. Debes, R. Heremans, and N. Frangiadakis, "Automatic fusion and classification of hyperspectral and LiDAR data using random forests," in *Proc. IEEE Geosci. Remote Sens. Symp.*, Jul. 2014, pp. 1245–1248.

[38] J. Li, Y. Ma, R. Song, B. Xi, D. Hong, and Q. Du, "A triplet semisupervised deep network for fusion classification of hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540513.

[39] S. K. Roy, A. Deria, D. Hong, B. Rasti, A. Plaza, and J. Chanussot, "Multimodal fusion transformer for remote sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5515620.

[40] S. Khan, M. Naseer, M. Hayat, S. W. Zamir, F. S. Khan, and M. Shah, "Transformers in vision: A survey," *ACM Comput. Surv.*, vol. 54, no. 10s, pp. 1–41, Jan. 2022.

[41] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[42] A. Amer Aleissaee et al., "Transformers in remote sensing: A survey," 2022, *arXiv:2209.01206*.

[43] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[44] Y. Zhou, X. Huang, X. Yang, J. Peng, and Y. Ban, "DCTN: Dual-branch convolutional transformer network with efficient interactive self-attention for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5508616.

[45] H. Li, B. Tu, B. Liu, J. Li, and A. Plaza, "Adaptive feature self-attention in spiking neural networks for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5500915.

[46] K. Ding, T. Lu, W. Fu, S. Li, and F. Ma, "Global–local transformer network for HSI and LiDAR data joint classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5541213.

[47] G. Zhao, Q. Ye, L. Sun, Z. Wu, C. Pan, and B. Jeon, "Joint classification of hyperspectral and LiDAR data using a hierarchical CNN and transformer," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500716.

[48] Y. Yu et al., "CapViT: Cross-context capsule vision transformers for land cover classification with airborne multispectral LiDAR data," *Int. J. Appl. Earth Observ. Geoinformation*, vol. 111, Jul. 2022, Art. no. 102837.

[49] Y. Zhang, Y. Peng, B. Tu, and Y. Liu, "Local information interaction transformer for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 16, pp. 1130–1143, 2023.

[50] M. Li, Y. Liu, G. Xue, Y. Huang, and G. Yang, "Exploring the relationship between center and neighborhoods: Central vector oriented self-similarity network for hyperspectral image classification," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 4, pp. 1979–1993, Apr. 2023.

[51] P. Xia, L. Zhang, and F. Li, "Learning similarity with cosine similarity ensemble," *Inf. Sci.*, vol. 307, pp. 39–52, Jun. 2015.

[52] T. Korenius, J. Laurikkala, and M. Juhola, "On principal component analysis, cosine and Euclidean measures in information retrieval," *Inf. Sci.*, vol. 177, no. 22, pp. 4893–4905, Nov. 2007.

[53] T. Mensink, J. Verbeek, F. Perronnin, and G. Csurka, "Distance-based image classification: Generalizing to new classes at near-zero cost," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 35, no. 11, pp. 2624–2637, Nov. 2013.

[54] L. Wang, Y. Zhang, and J. Feng, "On the Euclidean distance of images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 8, pp. 1334–1339, Jan. 2005.

[55] P. Ghamisi et al., "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.

[56] C. Debes et al., "Hyperspectral and LiDAR data fusion: Outcome of the 2013 GRSS data fusion contest," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2405–2418, Jun. 2014.

[57] P. Gader, A. Zare, R. Close, J. Aitken, and G. Tuell, "MUUFL Gulfport hyperspectral and LiDAR airborne data set," Univ. Florida, Gainesville, FL, USA, Tech. Rep. REP-2013-570, 2013. [Online]. Available: https://github.com/GatorSense/MUUFLGulfport

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[58] X. Du and A. Zare, "Scene label ground truth map for MUUFL Gulfport data set," Dept. Elect. Comput. Eng., Univ. Florida, Gainesville, FL, USA, Tech. Rep. 20170417, 2017.

[59] B. Rasti et al., "Feature extraction for hyperspectral imagery: The evolution from shallow to deep: Overview and toolbox," *IEEE Geosci. Remote Sens. Mag.*, vol. 8, no. 4, pp. 60–88, Dec. 2020.

[60] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," 2017, *arXiv:1711.05101*.

[61] P. Nabila and E. B. Setiawan, "Adam and AdamW optimization algorithm application on BERT model for hate speech detection on Twitter," in *Proc. Int. Conf. Data Sci. Appl. (ICoDSA)*, Jul. 2024, pp. 346–351.



Bokun Ma received the B.S. degree in information management and information systems from Xidian University, Xi'an, China, in 2023. He is currently pursuing the master's degree with Guangzhou Institute of Technology, Xidian University, Guangzhou, China.

His research interests include hyperspectral image processing and deep learning.



Caihong Mu (Senior Member, IEEE) received the B.S. degree in information engineering and the M.S. and Ph.D. degrees in circuit and system from Xidian University, Xi'an, China, in 2001, 2004, and 2010, respectively.

She is currently a Full Professor with the Key Laboratory of Intelligent Perception and Image Understanding of the Ministry of Education, Collaborative Innovation Center of Quantum Information of Shaanxi Province, School of Artificial Intelligence, Xidian University. Her research focuses on computational intelligence and machine learning with applications in hyperspectral image processing and recommender systems.



Yi Liu received the B.S. degree in information engineering, the M.S. degree in circuit and system, and the Ph.D. degree in communication and information systems from Xidian University, Xi'an, China, in 1999, 2002, and 2013, respectively.

Since 2002, he has been working as a Lecturer with Xidian University. His research interests include computational intelligence and image processing.



Xinyu He received the B.S. degree in automation engineering from Northeast Electric Power University, Jilin, China, in 2023. She is currently pursuing the master's degree with the School of Artificial Intelligence, Xidian University, Xi'an, China.

Her research interests include hyperspectral image processing and deep learning.



Mosa Haidarh received the B.S. degree in computer science from Taiz University, Taiz, Yemen, in 2009, and the M.S. degree in software engineering from Northeastern University, Shenyang, China, in 2022. He is currently pursuing the Ph.D. degree with the School of Artificial Intelligence, Xidian University, Xi'an, China.

His research focuses on hyperspectral image processing and deep learning.

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